

Unleashing International Trade through Financial Integration: Evidence from a Cross-Border Payment System

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Abstract

Using confidential data on South African exporters and the staggered rollout of a real-time gross settlement system across 14 African countries, we study how cross-border payment integration affects international trade. Adoption raises export volumes by 35% and transaction frequency by 45%, equivalent to a 6–8 percentage-point tariff reduction, with no evidence of trade diversion. Effects are concentrated in destinations with pre-existing domestic payment infrastructure and among firms with high external-finance-dependence. The system expands interbank activity, consistent with correspondent banking. Regional settlement systems can therefore promote trade and financial development while strengthening emerging markets' resilience to disruptions in global payment-network access amid rising geopolitical risk.

KEYWORDS: Financial integration, international trade, cross-border payments, payment speed

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1 Introduction

International trade payments are generally settled through banks linked by a global network of correspondent relationships concentrated in a small number of financial hubs. Despite the crucial role of settlement, the finance-and-trade literature has largely abstracted from this process, focusing instead on bank lending (Amiti and Weinstein, 2011; Paravisini, Rappoport, Schnabl, and Wolfenzon, 2015), trade finance provision (Niepmann and Schmidt-Eisenlohr, 2017a), and firms' external financing needs (Manova, Wei, and Zhang, 2015). Recent evidence highlights that international payment frictions matter: financial sanctions that sever banks' access to correspondent hubs significantly disrupt cross-border payments (Cipriani, Goldberg, and La Spada, 2023; Matvos and Neiman, 2026). Our paper bridges these two literatures by studying the real and financial effects of bypassing global payment hubs. While sanctions reveal the costs of losing access to established payment rails, we study the benefits of building alternative rails that circumvent those hubs.

Theory shows that payment contracts and trade finance allocate financing needs and counterparty risk across exporters, importers, and banks (e.g., Schmidt-Eisenlohr, 2013), while transaction-level evidence shows that payment terms respond to risk, enforcement, and financing conditions (Antras and Foley, 2015). However, we have little causal evidence on the real effects of the infrastructure through which banks settle these payments. Settlement infrastructure determines when foreign claims become final and usable, and how much operational and settlement risk banks and firms bear. It has also become a crucial tool for policymakers, as central banks and international organizations like the G7 and G20 have made faster and cheaper cross-border payments a priority for financial integration (Bindseil and Pantelopoulos, 2022).¹

We study this question using the staggered rollout of the Real-Time Gross Settlement (RTGS) system across the Southern African Development Community (SADC). The SADC-RTGS provides a direct shock to banks' cross-border settlement infrastructure by allowing participating banks to

¹The G7 Finance Ministers and Central Bank Governors communiqué of 2025 states: "Cross-border payments can have widespread benefits for citizens and economies worldwide. We remain committed to delivering cheaper, faster, more transparent, and more accessible cross-border payments."

settle cross-border transactions in real time through accounts centralized at the South African Reserve Bank. These accounts are pre-funded to reduce settlement risk, and the system provides immediate notifications for errors or unfunded accounts as well as real-time visibility of participant balances. The RTGS therefore changed both the speed and reliability of cross-border payment processing; it created a centralized settlement platform that combined faster settlement, lower counterparty risk, standardized procedures, and improved information flows.

The implementation of the SADC-RTGS system offers several empirical advantages for estimating the effect of international payment-system integration. First, the system launched in 2013 and was adopted in a staggered fashion, with all member countries eventually joining by the end of 2016. The implementation nature allows us to address common concerns about treatment timing and the selection of countries participating in the payment system. Second, there were no significant changes in regional trade agreements during the implementation window, which allows us to isolate the effect of payment integration from other changes in trade policy. Third, we provide evidence that the timing of entry into the RTGS is not predicted by origin, destination, or country-pair characteristics, including past trade between countries.

The SADC setting is particularly well suited to studying payment-settlement frictions because it combines substantial scope for payment integration with unusually rich microdata. At the start of the rollout, SWIFT reported that 48% of intra-African import/export settlement involved intermediation by a bank outside Africa, indicating considerable reliance on extra-regional payment chains (SWIFT, 2013). The SADC-RTGS created a regional alternative to these indirect settlement routes. At the same time, South Africa provides rare administrative data from a financially sophisticated emerging market. We link transaction-level customs records to annual balance-sheet information for the universe of South African exporters from 2010 to 2019, allowing us to observe export values and transaction frequency at the firm-destination level.

This combination of institutional variation and microdata allows us to test whether payment infrastructure affects real and financial outcomes. To discipline this test, we develop a framework in which payment integration affects exports by reducing an exporter-facing

payment-settlement friction. Building on trade models in which financial frictions raise the effective cost of exporting (Manova, 2013), the framework decomposes this friction into settlement delays, counterparty risk, and other bank-specific settlement costs. Settlement delays tie up working capital between shipment and receipt of usable funds, while counterparty risk captures the possibility that payments fail to settle as expected.

The framework yields three predictions that organize the empirical analysis. First, RTGS adoption should increase exports, especially to destinations with pre-existing domestic RTGS infrastructure, where cross-border integration can generate end-to-end speed gains. Second, effects should be stronger for firms with higher external financial dependence, because these firms are more exposed to the working-capital costs of settlement delays and payment uncertainty. Third, RTGS can expand interbank settlement activity. Because adoption entails a fixed cost, only banks with sufficiently large settlement volume in participating corridors adopt the system directly; the remaining banks access RTGS rails through adopter banks, which provide settlement services through correspondent relationships. The framework therefore predicts that banks with greater *ex ante* propensity to adopt the system expand interbank activity following RTGS activation. Such an expansion would be consistent with correspondent banking transmitting settlement improvements beyond direct participants and deepening financial integration.

The richness of the firm-level data enables us to isolate the causal effects of payment system integration across countries on real and financial outcomes, providing several empirical contributions. First, we document substantial baseline effects. After a destination joins the RTGS, firms increase export volume to that destination by 35% and the number of export transactions by 45%. These magnitudes are large but in line with evidence that financial frictions and payment technologies can have substantial real effects.² The effects are also consistent with our

²For comparison, related work documents large effects of financial and payment frictions. Michalski and Ors (2012) find that U.S. interstate banking integration increased domestic trade by 17–25%, while Matray et al. (2025) estimate a trade-financing multiplier of 4–5. Matvos and Neiman (2026) show that sanctions disrupt access to international payment networks, and Borchert et al. (2024) find that comparable disruptions to cross-border banking services reduce export participation by up to 35.2%. Relatedly, Barrot and Nanda (2020) and Higgins (2024) document large real effects of payment innovation.

framework predictions: RTGS reduces settlement delays and counterparty risk, lowering the payment-settlement friction faced by exporters.

Second, we provide evidence consistent with the payment-settlement mechanism by exploiting heterogeneity in domestic payment infrastructure. The framework predicts that the export response should be largest where domestic RTGS already exists, because cross-border RTGS generates end-to-end speed gains only when the destination's domestic payment leg can also settle in real time. Consistent with this prediction, we find that exports increase only in destinations with pre-existing domestic RTGS infrastructure. Destinations without domestic RTGS do not experience significant increases in exports in our preferred specification, suggesting that the domestic payment leg remains a bottleneck for settlement speed. This pattern is consistent with an end-to-end transaction speed channel rather than from other non-speed components of the settlement frictions.

Third, we test the framework's external-finance-dependence prediction. The model predicts that firms with greater exposure to working-capital needs should benefit more from reductions in settlement frictions, because the payment-settlement friction loads more heavily on their effective marginal cost of exporting. Consistent with this prediction, the export response is concentrated among firms in sectors with high external financial dependence: these firms increase total export volume after RTGS exposure, while less financially dependent firms do not respond significantly. Additional firm-level outcomes reinforce this interpretation: high-external-finance-dependence firms also experience higher total sales and larger receivables and payables, consistent with expanded trade activity and working-capital adjustment. Together, these results show that payment-system integration has real effects on firms by easing liquidity-related frictions where they should bind most.

Fourth, using bank-level data, we provide evidence on the correspondent-bank margin highlighted in our conceptual framework. Banks with headquarters or foreign subsidiaries in countries participating in the RTGS are more likely to adopt the system. These likely adopter banks also experience a significant increase in interbank assets after the countries in which they are lo-

cated join the RTGS. In the framework, adopter banks can provide RTGS settlement services to non-adopting respondent banks through correspondent relationships. The increase in interbank assets is therefore consistent with adopter banks expanding correspondent settlement activity after RTGS adoption, helping transmit the benefits of payment integration beyond firms whose own banks directly adopt the system and increasing financial integration.

We next assess whether RTGS creates new trade and whether its effects are amplified or attenuated by complementary channels. First, we find no evidence of trade diversion: firms exposed to RTGS do not reduce their export volume or transaction frequency with non-participant destinations after adoption. This suggests that payment-system integration creates new trade rather than merely reallocating exports across markets. Second, the effects are smaller in destinations with stronger pre-existing banking links to South Africa and are not explained by exchange-rate volatility. These patterns are consistent with RTGS generating the largest gains where prior cross-border banking connectivity was limited and payments were more likely to rely on longer, more indirect correspondent banking chains ([Matvos and Neiman, 2026](#)). Together, these findings provide further evidence that regional payment integration operates primarily by reducing payment-settlement frictions rather than through changes in exchange-rate-risk exposure.

Finally, using country-level trade flows and a gravity equation, we provide evidence of the aggregate consequences of countries participating in the RTGS. Trade between participant countries increases by around 35–49% on average. These results are robust to controlling for bilateral variables commonly used in gravity equation estimations, as well as fixed effects that account for the local demand and supply components. A back-of-the-envelope calculation reveals that the positive impact of payment integration on international trade is comparable to a reduction in tariffs by 6–8 percentage points.

Our paper relates to the broad literature on financial frictions, particularly studies that assess the impact of financial frictions on international trade ([Manova, 2013](#); [Schmidt-Eisenlohr, 2013](#); [Almeida et al., 2024](#)). Existing studies have shown that financial development shapes international trade primarily through the credit channel. For example, increased domestic finan-

cial development boosts trade flows (Beck, 2002; Contessi and De Nicola, 2012; Paravisini et al., 2023), and disruptions to credit and financial markets hinder both exports and imports (Amiti and Weinstein, 2011; Schnabl, 2012; Manova et al., 2015; Paravisini et al., 2015; Chaney, 2016; Niepmann and Schmidt-Eisenlohr, 2017a; Matray et al., 2025). A common thread in these studies is that financial frictions raise the variable costs of exporting disproportionately for constrained firms, as formalized by Manova (2013). Our distinctive contribution is identifying a specific *operational* friction—cross-border payment settlement speed and reliability—as a channel that is distinct from credit supply per se, and showing that it matters even after controlling for pre-existing banking connectivity and credit channels.

We also contribute to the literature on banking integration, payment networks, and international trade. Banking linkages and foreign-bank presence facilitate trade by reducing financial frictions (Caballero et al., 2018; Claessens and Van Horen, 2021). Recent work further shows that disruptions to cross-border banking and payment networks have substantial real effects. Borchert et al. (2024) find that the termination of correspondent relationships reduces exports by firms whose banks lose access to cross-border banking services, while Matvos and Neiman (2026) show that financial sanctions restrict banks’ access to major-currency payment networks and leave affected institutions reliant on longer and more fragile payment chains. Our paper differs by isolating payment settlement as a specific component of these broader financial relationships. The loss of a correspondent relationship can simultaneously affect payment execution, trade finance, currency clearing, and other banking services. By contrast, the SADC-RTGS directly changes the technology through which banks settle cross-border payments, without directly altering firms’ credit supply or trade-finance contracts. This setting allows us to identify payment-settlement frictions as a distinct financial channel affecting trade. Furthermore, we show that settlement integration expands interbank activity, consistent with correspondent banking transmitting settlement improvements beyond direct participants and deepening financial integration. More broadly, by creating regional settlement capacity and strengthening interbank links, such systems can increase emerging markets’ resilience to disruptions in access to global payment networks.

Our paper also contributes to the literature on payment-system innovation and its real effects. Existing work shows that domestic payment technologies affect firm and household outcomes through mobile money, faster government payments, cashless-payment adoption, and real-time settlement systems (Beck et al., 2018; Barrot and Nanda, 2020; Mariani et al., 2023; Higgins, 2024; D’Andrea and Limodio, 2024). We extend this literature from domestic payment technologies to cross-border settlement, where transactions span jurisdictions, banking networks, and working-capital cycles. We show that payment-system integration affects both real activity, through firm exports and balance sheets, and financial intermediation, through increased interbank settlement activity.

Taken together, our results identify payment settlement as a distinct financial friction. The paper shows that the technology banks use to settle cross-border payments affects firm exports, that these effects are concentrated where domestic payment infrastructure enables end-to-end speed gains, and that correspondent banking can transmit settlement improvements beyond direct adopters. These findings broaden the literature on finance and trade by showing that the real effects of financial integration operate not only through credit supply and banking relationships, but also through the payment infrastructure that determines the speed and certainty with which cross-border claims become usable liquidity. More broadly, our findings suggest that regional settlement integration can strengthen resilience to geopolitical disruptions in global payment networks.

2 Institutional Background

2.1 The SADC and its Real-Time Gross Settlement System

The Southern African Development Community (SADC) was established in 1992 and includes 16 member countries across southern Africa.³ The community created a framework for financial cooperation through the Finance and Investment Protocol, approved in 2006 and ratified by the

³Angola, Botswana, Comoros, Democratic Republic of Congo, Eswatini, Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia, Seychelles, South Africa, Tanzania, Zambia, and Zimbabwe.

Heads of State of all original member countries. This treaty specifically encourages cooperation among central banks for payment, clearing, and settlement systems. Seychelles joined in 2008, and the Union of the Comoros joined in 2018. Our analyses consider only the SADC members as of 2013, when the payment system implementation began, excluding Comoros and Madagascar (suspended between 2009 and 2014 due to political instability).

A major advancement in the region's financial integration was the development of the Real-Time Gross Settlement (RTGS) system. In October 2010, the Committee of Central Bank Governors approved a proof-of-concept in the Common Monetary Area, comprising Lesotho, Namibia, South Africa, and Eswatini (formerly Swaziland). The system officially launched in July 2013. After this pilot phase, implementation expanded to other countries in stages, as detailed in [Table 1](#). This electronic settlement system enables either immediate or scheduled settlement of payments between participating banks. The system operates in South African rand (ZAR) during business hours (8 AM to 5 PM South African time) and is managed by the South African Reserve Bank on behalf of all participating central banks.

[INSERT [TABLE 1](#) ABOUT HERE]

The payment system functions as a central electronic platform that processes rand-denominated payments to facilitate cross-border trade within the region. Participating banks can transfer funds and settle transactions across SADC countries immediately and on a gross basis, meaning each payment is settled individually. This design reduces banking costs by eliminating the need for intermediary correspondent banks. Participant banks can exchange payments directly, allowing them to centralize funds in a single account rather than maintaining separate accounts at different banks. The system also speeds up access to funds for businesses and other bank clients by reducing processing time.

Commercial and central banks participating in the system must pre-fund their accounts, which reduces settlement risks for transfers within the SADC. This pre-funding mechanism provides greater security for the 82 banks connected to the system. The platform has evolved from han-

dling only high-value urgent payments to including scheduled payments, with plans to add batch processing for low-value payments in the future. The system follows principles focused on facilitating regional transactions, using existing infrastructure, adhering to international standards, and enabling straight-through processing.

The regional settlement system offers multiple capabilities, including electronic funds transfers, card and ATM transaction clearing, and settlement for cross-border securities transactions. Participants benefit from the system's ability to process payments that are immediate, final, and irrevocable. The system also reduces costs through standardized processes, real-time notifications of errors or unfunded accounts, and a web dashboard that shows participants' balances in real time.

For businesses and individuals in the SADC, the payment system has several advantages. It provides faster payment services, simplifies cross-border transactions, reduces the need for cash, and improves security. It also creates more transparent pricing and standardized payment methods. Users gain the flexibility to compare services and choose between immediate cash-out options or waiting for favorable exchange rates. Overall, the system represents a significant achievement in regional cooperation, promoting economic integration and delivering concrete benefits to participating countries and their financial institutions.

2.2 The SADC and International Trade Promotion

When examining how the RTGS system affected international trade, we must consider whether other simultaneous initiatives within SADC might have influenced trade patterns. The community has implemented various protocols to enhance regional cooperation. In 2008, the SADC established a Free Trade Area to promote regional trade, which was fully implemented by 2012 and includes 13 member countries (all except Angola, Comoros, and the Democratic Republic of Congo). Also in 2008, the Community agreed to join the Common Market for Eastern and Southern Africa and the East African Community to form the Tripartite Free Trade Area, also called the

African Free Trade Zone. In 2015, member countries signed an agreement to establish this larger free trade area, though it has not yet been fully implemented.

While these trade liberalization initiatives represent important advancements to the region's economic integration, they were already in place before the payment system roll-out began. As such, these advancements in trade agreements do not present an identification threat to our current analysis. Our empirical strategy leverages the staggered timing of RTGS adoption across countries between 2013 and 2016, while controlling for time-invariant bilateral trade relationships through firm-destination-cohort fixed effects. This approach ensures that any estimated effects capture the impact of payment system integration rather than pre-existing trade agreements or other regional integration initiatives.⁴

3 Conceptual Framework

This section presents a framework to guide our empirical analysis. In our setting, cross-border payment integration affects exports by reducing the payment-settlement friction faced by exporters. This friction reflects three related components: settlement delays, counterparty risk, and the bank-specific settlement cost. Settlement delays create liquidity costs because exporters must produce and ship goods before receiving payment from foreign buyers. Counterparty risk captures the possibility that payments do not settle as expected because of insufficient balances, correspondent-bank frictions, operational failures, disputes, or information frictions. The pre-funded nature of RTGS directly reduces this risk by requiring participating banks to hold balances before payments are processed. Finally, banks differ in their settlement cost across corridors, reflecting heterogeneity in operational capacity, correspondent networks, compliance systems, treasury operations, and experience processing payments in a given corridor. These bank-corridor costs are invariant to RTGS access: RTGS changes the settlement technology, not the underlying efficiency of a given bank in a given corridor. The premise that conditions at a firm's own bank trans-

⁴Moreover, as we show in Internet Appendix [Table IA.3](#), observable country characteristics (e.g, trade flows and economic indicators) do not predict the timing of RTGS entry.

mit to that firm's exports — because firm-bank relationships are sticky and cannot be costlessly re-optimized — is well documented empirically (Amiti and Weinstein, 2011; Paravisini et al., 2015).

Consider firm i in sector s exporting to destination j . Demand in destination j is CES:

$$q_{ijs} = A_j p_{ijs}^{-\sigma}, \quad \sigma > 1, \quad (1)$$

where A_j is the destination demand shifter and σ is the elasticity of substitution. The unit cost of producing and delivering the good is C_{ijs} . Let a fraction $d_s \in (0, 1)$ of this cost be exposed to settlement frictions. The parameter d_s captures external financial dependence: firms with higher d_s rely more on external liquidity to cover production and shipping costs before foreign payments are received. In modeling frictions as a sector-specific multiplicative cost shifter on export costs, and in exploiting the interaction between country-level frictions and sectoral external financial dependence, we follow the trade-and-finance literature (Manova, 2013; Chor and Manova, 2012). Our innovation relative to that literature relies on the nature of the friction — settlement delays, counterparty risk, and bank-corridor costs rather than credit and enforcement frictions — and, in the focus on the settlement mechanism and its interbank transmission. For simplicity, we abstract from firm heterogeneity and selection into exporting, so the predictions below concern the intensive margin of trade.

Let $\omega_{bj} \geq 0$ denote the payment-settlement friction associated with exporting through bank b to destination j , expressed as a proportional markup on the exposed share of export costs; its nonnegativity follows from the decomposition below, whose components are all nonnegative. The exporter's effective marginal cost is

$$C_{ibjs}^e = C_{ijs}(1 + d_s \omega_{bj}). \quad (2)$$

Under CES demand, export revenues are

$$X_{ibjs} = A_j \left[\frac{\sigma - 1}{\sigma} \frac{1}{C_{ijs}(1 + d_s \omega_{bj})} \right]^{\sigma - 1}, \quad (3)$$

so that, since the friction is a continuous variable, exports are strictly decreasing in it:

$$\frac{\partial \log X_{ibjs}}{\partial \omega_{bj}} = -\frac{(\sigma - 1)d_s}{1 + d_s \omega_{bj}} < 0. \quad (4)$$

How does RTGS access affect this friction? To organize the channels, consider the path of a cross-border payment. The payment travels an end-to-end chain: it is originated by the importer at its bank in the destination country, routed through that country's domestic payment system to a bank with cross-border settlement access, settled between banks across the border, and finally credited to the exporter by its bank. We decompose the friction according to which link of this chain generates each cost and, therefore, which part of the payment infrastructure each cost responds to. End-to-end settlement speed is governed by the slowest link in the chain: cross-border RTGS accelerates the interbank leg, but the payment reaches that leg promptly only if the destination's domestic leg also settles in real time, and the exporter is not paid until every link has cleared. The delay component of the friction therefore depends on both the bank's cross-border RTGS access and the destination's domestic payment infrastructure. Counterparty risk, by contrast, is resolved on the cross-border leg itself: because participating banks must hold balances before payments are processed, the pre-funded design of RTGS reduces the risk that payments fail to settle, independently of what happens downstream. The risk component therefore depends on cross-border RTGS access alone. Finally, whichever settlement technology is used, a bank must execute the settlement, and banks may differ in how efficiently they do so in a given corridor. This third component is a characteristic of the bank-corridor pair, and we assume it is orthogonal to the available settlement technology.

Formally, let $z_{bj} \in \{0, 1\}$ indicate access to cross-border RTGS settlement in corridor j , either directly or through a correspondent bank, and let $D_j \in \{0, 1\}$ indicate that destination j has domestic RTGS infrastructure. We can decompose the payment-settlement friction as

$$\omega_{bj}(z_{bj}, D_j) = \underbrace{T_j(z_{bj}, D_j)}_{\text{settlement delay}} + \underbrace{\delta_j(z_{bj})}_{\text{counterparty risk}} + \underbrace{c_{bj}}_{\text{bank-corridor cost}}. \quad (5)$$

To simplify notation, each component is expressed directly in the same cost units as the friction, absorbing any scaling parameter into the respective term. Every component is nonnegative ($T_j, \delta_j, c_{bj} \geq 0$), so the friction is nonnegative under every settlement route considered below.

The two properties of RTGS settlement described above suffice to characterize how the friction responds to RTGS access and, because access is binary, both are statements about discrete differences. First, reflecting the bottleneck role of domestic payment infrastructure, cross-border RTGS access reduces settlement delays if, and only if, the destination has domestic real-time settlement:

$$\lambda_j \equiv T_j(0, D_j) - T_j(1, D_j), \quad \lambda_j > 0 \text{ if } D_j = 1, \quad \lambda_j = 0 \text{ if } D_j = 0. \quad (6)$$

The parameter λ_j represents the reduction in settlement delay that any bank in corridor j obtains upon gaining cross-border RTGS access. Second, reflecting the pre-funded design of RTGS, access weakly reduces counterparty risk regardless of the destination's domestic infrastructure:

$$\eta_j \equiv \delta_j(0) - \delta_j(1) \geq 0. \quad (7)$$

The parameter η_j is, therefore, the reduction in counterparty risk from settling through the pre-funded system rather than the alternative settlement arrangement in corridor j . The third component, c_{bj} , reflects heterogeneity in operational capacity, correspondent networks, compliance systems, treasury operations, and experience processing payments in corridor j . It is a characteristic of the bank-corridor pair and is invariant to RTGS access: RTGS changes the corridor-level

speed and risk components, while c_{bj} captures persistent heterogeneity in banks' ability to process payments in corridor j . This heterogeneity generates persistent differences in the friction faced by otherwise similar exporters, depending on the bank through which they settle.

Taking the discrete difference of (5) between $z_{bj} = 1$ and $z_{bj} = 0$:

$$\Delta_z \omega_{bj} \equiv \omega_{bj}(1, D_j) - \omega_{bj}(0, D_j) = -(\lambda_j + \eta_j) \equiv -\Gamma_j \leq 0, \quad (8)$$

with strict inequality if and only if at least one channel is active in corridor j : the speed channel requires $D_j = 1$, and the risk channel requires $\eta_j > 0$. Writing $\omega_{bj}^0 \equiv \omega_{bj}(0, D_j)$ for the baseline friction without cross-border RTGS access, and combining (8) with (3), RTGS access increases exports by

$$\Delta_z \log X_{ibjs} = (\sigma - 1) \log \left[\frac{1 + d_s \omega_{bj}^0}{1 + d_s (\omega_{bj}^0 - \Gamma_j)} \right] \geq 0, \quad (9)$$

with strict inequality if and only if $\Gamma_j > 0$. The magnitude of the response depends on the baseline friction ω_{bj}^0 at which the discrete change is evaluated, so cross-destination comparisons below hold the baseline friction fixed explicitly. The framework delivers three testable predictions.

Prediction 1: RTGS increases exports, especially when domestic RTGS is present. Compare two destinations that are identical in all respects relevant for the export response — in particular, they share the same baseline friction ω^0 and the same counterparty-risk reduction η — and differ only in domestic RTGS infrastructure. For the destination with domestic RTGS ($D_j = 1$), both channels operate, with speed gain $\lambda \equiv \lambda_j > 0$ by (6):

$$\Delta_z \log X_{ibjs} \Big|_{D_j=1} = (\sigma - 1) \log \left[\frac{1 + d_s \omega^0}{1 + d_s (\omega^0 - \lambda - \eta)} \right] > 0. \quad (10)$$

For the destination without domestic RTGS ($D_j = 0$), the speed gain is zero by (6), since the domestic leg remains a bottleneck, and only the risk channel operates:

$$\Delta_z \log X_{ibjs} \Big|_{D_j=0} = (\sigma - 1) \log \left[\frac{1 + d_s \omega^0}{1 + d_s (\omega^0 - \eta)} \right] \geq 0, \quad (11)$$

with strict inequality if and only if $\eta > 0$. Destinations without domestic RTGS therefore help separate non-speed settlement benefits from end-to-end speed gains. Holding the baseline friction fixed, the terms in $\log(1 + d_s \omega^0)$ cancel, and the difference between the two responses captures the additional effect of end-to-end speed gains:

$$\Delta_z \log X_{ibjs} \Big|_{D_j=1} - \Delta_z \log X_{ibjs} \Big|_{D_j=0} = (\sigma - 1) \log \left[\frac{1 + d_s (\omega^0 - \eta)}{1 + d_s (\omega^0 - \lambda - \eta)} \right] > 0, \quad (12)$$

where the strict inequality follows from $\lambda > 0$.

Prediction 2: Effects are stronger for firms with higher external financial dependence. The export response to RTGS access in (9) is strictly increasing in d_s whenever $\Gamma_j > 0$:

$$\frac{\partial}{\partial d_s} \Delta_z \log X_{ibjs} = (\sigma - 1) \left[\frac{\omega_{bj}^0}{1 + d_s \omega_{bj}^0} - \frac{\omega_{bj}^0 - \Gamma_j}{1 + d_s (\omega_{bj}^0 - \Gamma_j)} \right] > 0, \quad (13)$$

because $\omega / (1 + d_s \omega)$ is strictly increasing in ω . A given reduction in settlement frictions therefore generates larger export gains for firms with higher external financial dependence.

Prediction 3: RTGS expands interbank (correspondent) settlement activity and transmits benefits beyond direct adopters. The benefits of RTGS need not be limited to exporters whose own banks join the system directly. When a country joins the RTGS, each bank operating there chooses how to access the new settlement technology: adopt it directly, or settle through a bank that does. Adoption carries a fixed cost $F > 0$ — participation and connectivity requirements, compliance systems, and the treasury operations needed to maintain pre-funded balances. The alternative

is the correspondent route: a non-adopting bank routes its cross-border payments through an adopter bank at a per-transaction fee ϕ . We take the fee as given, with $0 < \phi < \Gamma$, where, to ease notation, we suppress corridor heterogeneity for this prediction and write Γ for the per-transaction friction reduction on RTGS rails. These bounds are participation constraints: a fee below the friction reduction is required for any bank to use the correspondent route, and any positive fee makes serving correspondents profitable for adopters, whose marginal cost of processing additional payments on rails they already operate is negligible. Where the fee falls within this range depends on competition among adopter banks.

Let V_b denote bank b 's RTGS-relevant settlement volume (the cross-border payments it settles in corridors covered by the system), which should be larger for bigger banks and for banks with a regional footprint, such as headquarters or subsidiaries in participating countries.⁵ Relative to the status quo, adopting yields settlement-cost savings ΓV_b net of the fixed cost F , while the correspondent route yields $(\Gamma - \phi)V_b$ at no fixed cost. Since $(\Gamma - \phi)V_b > 0$, every bank prefers some form of RTGS access to the status quo. The adoption margin compares the fee savings from direct access with the fixed cost: bank b adopts if and only if

$$\Gamma V_b - F \geq (\Gamma - \phi)V_b \iff V_b \geq V^* \equiv \frac{F}{\phi}. \quad (14)$$

The cutoff delivers three results. First, selection into adoption: only banks with sufficiently large RTGS-relevant settlement volume find direct adoption profitable. Second, interbank settlement expands: banks below the cutoff route their RTGS-corridor payments through adopter banks, so adopters settle on behalf of non-adopters, and correspondent settlement activity increases by the total volume of the banks below the cutoff. Third, transmission: assuming non-adopting banks price settlement services to their exporter clients at cost, exporters served by these banks obtain a friction reduction of $\Gamma - \phi > 0$, so the benefits of settlement integration extend beyond

⁵We treat V_b as invariant to the bank's choice between direct adoption and the correspondent route. In Appendix [IA.1.6.5](#) we relax this assumption and let settlement volume under direct adoption equal γV_b with $\gamma \geq 1$, reflecting the additional settlement business induced by the larger friction reduction on RTGS rails. The adoption margin remains a cutoff rule with a weakly lower cutoff, $F/[\phi + (\gamma - 1)\Gamma]$, and all takeaways are unchanged.

direct participants. The magnitude of the transmission is an equilibrium object: competition among adopter banks pushes ϕ toward cost, enlarging the pass-through, while a correspondent with market power prices close to Γ , shrinking it.

This mechanism delivers the bank-level tests we take to the data. Banks with larger RTGS-relevant settlement volume — banks with headquarters or foreign subsidiaries in participating countries — should be more likely to adopt the system, and adopter banks should expand correspondent settlement activity after the country in which they operate joins the system. Formal statements and derivations are provided in Appendix [IA.1](#).

4 Data and Sample Construction

4.1 Data Sources

Firm-Level Data. One of the major challenges in estimating the effects of financial integration on international trade is the limited availability of detailed, firm-level data that covers exporters and destinations. We address this by using administrative firm-level tax data from the South African Revenue Service for the 2010–2019 period. The primary source is the South African Corporate Income Tax data, collected annually for the tax year ending in February. Firms must submit corporate income tax returns with self-reported information about their income, expenditures, equity, liabilities, capital items, and tax credits. Almost all reporting items are compulsory, and compliance is high since the revenue service may audit firms in any given year. We use this data for our firm-level analysis of total exports, examining variables such as total sales, value added, receivables, payables, and labor costs in our tests of how RTGS exposure affects firm performance.

We complement this information with firm employment data from employee income tax certificates (IRP5 forms). We aggregate these employment records by Pay-As-You-Earn reference number and link them to the corresponding firm’s tax reference number. Companies identified by a unique tax reference number may have multiple employment reference numbers, so we match all employees with matching reference numbers to their corresponding firms ([Pieterse et al., 2018](#)).

A crucial feature of the South African Revenue Service and National Treasury Firm-Level Panel in our paper is the detailed firm-level customs records. The data covers 2010 to 2019 and includes transaction-level information on export value, destination, product, and transaction date. We use this customs data for our primary firm-destination-level analysis, creating measures of export volume (in South African rand, ZAR) and number of export transactions as our main dependent variables. This allows us to build a firm panel dataset with the total value of exports and the number of transactions performed with each destination country in any given year by product. We also use this data to construct our extensive margin measures, including the number of destination countries and the number of destination-by-product combinations at both the HS 4-digit and 6-digit levels.

Aggregate Data. Our second data source comes from the gravity database compiled by the *Centre d'Etudes Prospectives et d'Informations Internationales* (Conte et al., 2022). This dataset contains international trade information from more than 200 countries and 5,000 products between 1994 and 2022. It comes from reconciled data reported by almost 150 countries to the United Nations Statistics Division (United Nations, 2023). We use these bilateral trade flows for our country-level gravity estimates to examine the aggregate effects of RTGS participation on trade between member countries. The dataset also contains information for origin–destination country pairs in each year, including geographic, cultural, trade facilitation, and macroeconomic variables. We use these bilateral controls in our gravity specifications, including geographical distance, shared borders, free trade agreements, customs unions, and measures of contiguity. Additionally, we use this data to construct our export demand controls for firm-level specifications, estimating destination-specific demand shocks by regressing bilateral trade flows on importer–year fixed effects while excluding South African trade flows.⁶

⁶See Internet Appendix IA.3 for details on how we estimate export demand. The intuition is as follows: if a country experiences a positive demand shock (e.g., economic growth, favorable policy changes), this should increase its imports from *all* trading partners, not just South Africa. Our approach captures this common demand component that affects all exporters to the same destination in a given year. Therefore, our estimated destination-specific demand shocks are orthogonal to bilateral factors (such as RTGS adoption) that affect only specific country pairs.

Bank-Level Data. Finally, we also use bank-level data from BankFocus in two empirical tests. We begin with the international network of South African banks, extracting information on all South African banks, their subsidiaries, and parent companies. We then identify 75 South African banks with more than 4,000 subsidiaries/parents in 128 countries. We complement this with information on each bank’s total assets before the payment system implementation in 2012. This information comes from regulatory forms that the South African Reserve Bank requires all commercial and mutual banks to complete monthly. We then use information from all international bank groups with at least one subsidiary in the African Continent to highlight the determinants of payment system adoption and the implications for interbank activity.

4.2 Empirical Strategy

A major challenge in studying the causal effect of increased financial integration on international trade is that both processes often happen simultaneously, making it difficult to isolate the causal effect. To address this challenge, our empirical strategy uses the staggered implementation of the payment system to understand the effects of increased financial integration on international trade. Recent research has highlighted that traditional two-way fixed effects models can produce misleading results when treatments occur at different times.⁷

Stacked Regression Estimation. Given the staggered implementation of the payment system, we stack implementation cohort-specific data to calculate an average effect across all participants to overcome the problems highlighted above (Gormley and Matsa, 2011; Cengiz et al., 2019; Baker et al., 2022).⁸ Although our empirical strategy does not require that the treated and control groups be similar, we use a matching procedure to find a control group similar to the treated units.⁹

⁷See, e.g., Goodman-Bacon (2021); Callaway and Sant’Anna (2021); de Chaisemartin and d’Haultfœuille (2020). Goodman-Bacon (2021) shows that the treatment effects using this specification are a combination of the weighted average of the unit-time treatment effects. In the context of two-way fixed effects, these treatment effect coefficients can be potentially weighted by negative weights (de Chaisemartin and d’Haultfœuille, 2020).

⁸This specification uses stricter criteria for the choice of the control groups. By aligning events by the event time, we prevent the negative weighting of some events that may occur with a staggered design.

⁹Results for the non-matched sample are quantitatively similar to our baseline results (see Table IA.5).

We construct cohorts by RTGS implementation year in each country and stack them to estimate the average treatment effect. We use an estimation window of 3 years around each cohort year. We do so because export data start in 2010 (3 years before the 2013 cohort was treated) and to exclude the period after the COVID-19 pandemic started. Since all SADC participant countries eventually joined the payment system, this choice is not endogenously determined in our setting. Moreover, we provide evidence in [Table IA.3](#) in the Internet Appendix that pair-specific and country-specific characteristics do not predict entry into the RTGS.

Firm-Level Matching. To enhance the credibility of our identification strategy, we use Coarsened Exact Matching (CEM) to construct a balanced control group ([Iacus et al., 2011](#)). The matching approach offers several advantages: it ensures that matched samples satisfy the congruence principle (that is, treated and control groups have similar empirical distributions of confounding variables), reduces model dependence, and provides bounds on the degree of imbalance between groups.¹⁰

Our choice of matching variables is guided by trade and finance theory. First, we match on firm sector (2-digit ISIC classification) to capture systematic differences in technological intensity and financial vulnerability across industries. [Manova \(2013\)](#) shows that sectors requiring higher upfront investment or with fewer collateralizable assets are more exposed to financial frictions, making the sector a crucial balancing dimension. Second, we include firm size (proxied by total sales, coarsened into quartiles) since larger firms are typically less credit-constrained and more likely to overcome the fixed costs of entering export markets ([Bernard et al., 2003](#)). Third, we match on total factor productivity (TFP, coarsened into quartiles). In heterogeneous-firm models of trade, more productive firms are more likely to become exporters and earn higher revenues, conditional on exporting ([Melitz, 2003](#); [Manova, 2008](#)). Balancing on productivity, therefore, ensures that treated and control groups do not differ systematically in their underlying export potential.

¹⁰Unlike propensity score matching, CEM does not require iterative procedures or balance checking, and it automatically restricts the data to areas of common empirical support ([Blackwell et al., 2009](#)). This is particularly important in our setting, where firms self-select into exporting to different destinations.

Our treated group consists of firms that exported to countries that eventually joined the RTGS system, while our control group includes matched firms that exported only to non-participating countries in 2012. This matching approach ensures that treated and control firms are similar across observable dimensions that could influence export performance, while preserving the natural randomness inherent in the staggered rollout of the payment system.

[INSERT [TABLE 2](#) ABOUT HERE]

In [Table 2](#), we show that our matching strategy successfully balances the treated and control groups across a comprehensive set of firm characteristics. The table reports summary statistics for our matched sample, showing that firms exporting to payment system participant countries are very similar to matched firms exporting solely to non-participant trade partners. Column (4) shows that all mean difference tests display statistically insignificant p -values (corrected using [Clarke, Romano, and Wolf’s \(2020\)](#) adjustment for multiple hypothesis testing), confirming that our matching approach selects a control group that resembles treated firms across a variety of characteristics, including financial variables (loans, receivables, payables), operational measures (capital stock, value added), and export intensity.

5 Firm-Level Evidence

5.1 Faster Cross-Border Payments and Firm Exports

Our empirical strategy uses a difference-in-differences analysis, leveraging the gradual implementation of the payment system across countries. To strengthen our approach, we implement a matching technique described above to obtain a more balanced sample. The control group is the sample of matched exporting firms of similar size, productivity levels, and sector that exported only to non-participating countries in 2012. This approach helps us estimate the treatment effects cleanly and enhances the reliability of our analysis.

Specifically, we estimate the following baseline model:

$$T_{i,j,c,t} = \exp \left\{ \alpha_{i,j,c} + \alpha_{c,t} + \beta_{RTGS} \cdot [RTGS_{j,c} \times Post_{c,t}] + \beta_d \cdot Export\ Demand_{j,t} \right\} \cdot \epsilon_{i,j,c,t} \quad (15)$$

in which i represents the firm, j the destination country, c cohort, and t year. $T_{i,j,c,t}$ is our firm-level outcomes for international trade: (i) the total volume exported (denominated in South African rand, ZAR), to country j in year t ; or (ii) the total number of export transactions that firm i had with country j in year t . $\alpha_{i,j,c}$ are firm-destination-cohort fixed effects, $\alpha_{c,t}$ are cohort-year fixed effects. The firm-destination-cohort fixed effects control for confounders specific to each firm and trade partner that do not vary in our sample period (e.g., any geographical or cultural connection between the firm and the destination country). Moreover, we control for the destination-country export demand ($Export\ Demand_{j,t}$), as detailed in Appendix [IA.3](#). $RTGS_{j,c}$ is the binary variable that equals one for the countries that participated in the RTGS in cohort c , and $Post_{c,t}$ is the binary variable that equals one for years after cohort c 's entrance into the system. Our coefficient of interest is β_{RTGS} , representing the average effect of the trade partner j 's inclusion in the payment system on firm i 's exports.

Finally, to account for heteroskedasticity in trade data and the presence of zeros in the dependent variables when analyzing export volume and the number of transactions, we use the Poisson Pseudo-Maximum Likelihood Estimator proposed by [Santos Silva and Tenreiro \(2006\)](#).

[INSERT [TABLE 3](#) ABOUT HERE]

[Table 3](#) shows the results estimating [Equation \(15\)](#). Columns (1) and (2) show the results when using the firm's export volume as the dependent variable, while Columns (3) and (4) use the number of transactions as the dependent variable. Our results show that firms exporting to countries participating in the SADC-RTGS increase their export volume (35–38.5%) and their number of export transactions (45–47%), on average.¹¹ These findings

¹¹The elasticities are calculated as $\exp(\beta) - 1$, from the estimates in [Table 3](#).

suggest that faster and more reliable cross-border payment systems substantially facilitate international trade by reducing transaction costs and uncertainty.

Our results are robust to controlling for firm-destination-cohort fixed effects, suggesting these effects are not driven by firm characteristics that might affect exports to a specific destination. Notably, these results are not driven by destination country demand or other time-varying shocks that might affect export activity, such as an increase in credit supply caused by the payment system (see, e.g., [D’Andrea and Limodio \(2024\)](#)).

To assess the plausibility of our estimated effects, we benchmark them against other trade facilitation measures documented in the literature. Our finding of a 35% increase in export volume is substantial, but consistent with the magnitude of effects found for other frictions in cross-border commerce. The most directly comparable benchmark is [Michalski and Ors \(2012\)](#). The authors study domestic banking integration in the United States from the late 1970s through the mid-1990s. They show that banking integration across states increased domestic (interstate) trade by 17–25%. Several factors explain why our effects are expected to be larger. First, we examine *international* rather than domestic trade, where frictions are substantially more restrictive ([Schmidt-Eisenlohr, 2013](#)). Second, our context focuses on emerging-market exporters, where financial constraints are more binding ([Almeida et al., 2024](#)) and improvements in payment infrastructure yield disproportionately larger gains. Third, recent studies also support the magnitude of our effects by documenting significant impacts from improvements in payment speed. [Barrot and Nanda \(2020\)](#) and [Higgins \(2024\)](#) highlight substantial real effects from faster payments and cashless adoption, while our cross-border payment integration represents an even more dramatic improvement—from multi-day correspondent banking to real-time settlement.

Recent studies document larger or similar impacts: [Matray et al. \(2025\)](#) show that the shutdown of the U.S. Export–Import Bank reduced exports with a financing multiplier of 4–5. [Borchert et al. \(2024\)](#) find that banking shocks strongly propagate into trade, with financially exposed exporters experiencing a large decline of up to 35.2% in the likelihood of exporting four years after the shock, further reinforcing the plausibility of our findings.

[INSERT [FIGURE 1](#) ABOUT HERE]

One potential concern with our empirical approach is whether our control units are an appropriate comparison group for the firms exposed to the payment system. In [Figure 1](#), we provide evidence suggesting that, before the implementation of the payment system, the treated and non-treated firms behaved similarly in terms of their export volume and transaction number. In other words, firms that did not export to countries participating in the SADC-RTGS behaved similarly to exporters exposed to the system in the years before its implementation. The figure also shows evidence of divergence in volume and transactions in the years following the implementation of SADC-RTGS, which provides reassuring evidence for our empirical strategy.

Our baseline results are robust to alternative control groups. [Table IA.5](#) in the Internet Appendix shows that the results hold when using the non-matched sample. Additionally, [Table IA.6](#) demonstrates that our findings remain stable when restricting the control group to firms exporting only to other African countries that do not participate in the RTGS, addressing potential concerns about unobserved heterogeneity across global trade destinations.

Trade Diversion versus Trade Creation. Since we find that participation in the international payment system increases a firm’s exports, one natural question is whether this increase is due to trade flows being diverted from countries outside the SADC to participant countries. If firms are just switching the destination of their exports and not generating new trade flows, this could raise concerns about the global welfare implications of integrating payment systems. This problem, often referred to in the trade literature as *trade diversion*, has been extensively examined in the case of trade liberalization policies ([Viner, 1950](#); [Dai et al., 2014](#)).

[INSERT [TABLE 4](#) ABOUT HERE]

In [Table 4](#), we test whether the cross-border payment system influences trade creation or merely shifts trade away from one country toward payment-system participant countries. We test the hypothesis of trade diversion by including in [Equation \(15\)](#) one additional variable that

captures the effect that the adoption of the RTGS may have caused on exports to countries outside the payment system, which we refer to as *RTGS Firms Spillover*. This is a binary variable that takes the value 1 when the destination country is not an RTGS member, but the firm is directly affected by RTGS adoption by exporting to at least one participating country after adoption. In other words, this variable captures the marginal effect of RTGS adoption on a firm's trade volume and transactions with countries outside the payment system.

We find no evidence of an impact on the volume of exports or the number of transactions from firms that export to SADC-RTGS participant countries to other trade partners that do not participate in the system after the system launch.¹² These results suggest that facilitating cross-border payments contributes to trade creation and not to trade diversion at the firm level.

The Role of Domestic Payment Infrastructure. To validate our identification strategy and pin down the effects of faster payments from other possible channels (as we discuss in [Section 3](#)), we exploit the fact that only some destinations enjoyed real-time settlement stemming from cross-border payment system integration. If our identification strategy is valid, then the pre-existence of domestic RTGS infrastructure should be a relevant factor in making payments faster, as it represents the necessary pre-existing infrastructure at the destination to enable true real-time settlement. To test for this, we exploit the fact that some destinations had the required domestic infrastructure to settle payments instantly, while others did not. Therefore, we include in [Equation \(15\)](#) a triple interaction term between the pre-existence of a domestic RTGS indicator variable and the indicator variables for treatment and post-adoption of the SADC-RTGS.

[INSERT [TABLE 5](#) ABOUT HERE]

[Table 5](#) presents the results. Consistent with our framework's settlement delay channel, the export response is concentrated in destinations with pre-existing domestic real-time settlement infrastructure. In our preferred specification, destinations without domestic RTGS experience

¹²This finding also lessens concerns that the effects we observe are due to, e.g., greater firm productivity.

no statistically significant increase in exports. This pattern is consistent with domestic payment infrastructure being necessary for cross-border RTGS integration to generate end-to-end speed gains: when the domestic settlement leg remains slow, faster cross-border settlement need not translate into substantially faster receipt of usable funds by exporters or importers. The results therefore support the prediction that the trade effects of RTGS integration should be strongest where domestic infrastructure allows improvements in cross-border settlement speed to propagate through the entire payment chain.

5.2 Inspecting the Mechanisms

Having established that the RTGS system increases firm-destination exports, we now examine the mechanisms through which payment integration affects firms' *total* export performance across all destinations. Guided by the framework in [Section 3](#), we focus first on the financial-frictions channel: RTGS should reduce the settlement frictions created by the time and uncertainty between shipment and receipt of usable cash. We then examine two alternative or complementary channels—pre-existing bank linkages and exchange-rate risk.

5.2.1 Financial Frictions and Working-Capital Needs

A central prediction of the framework is that RTGS integration should matter most for firms whose export activity is more exposed to working-capital needs. International trade is capital intensive: firms must finance production, inventory, shipping, and documentation before receiving payment from foreign buyers ([Manova, 2013](#); [Schmidt-Eisenlohr, 2013](#)). When payment settlement is slow or uncertain, this cash-conversion cycle ties up working capital and increases the financing burden of exporting. By accelerating settlement and increasing the certainty with which foreign revenues become usable liquidity, the RTGS should therefore disproportionately benefit firms for which external finance is more important.

To test this prediction, we construct a firm-level measure of exposure to the RTGS based on pre-existing export patterns. The measure captures the share of a firm's baseline exports shipped to destinations that have adopted the payment system by year t :

$$RTGS\ Exp_{i,t} = \sum_{j=1}^n \left[\frac{Exports_{i,j,2012}}{Total\ Exports_{i,2012}} \right] \cdot [RTGS_{j,c} \times Post_{c,t}], \quad (16)$$

where i denotes the firm, j the destination country, c the adoption cohort, and t the year. The first term is firm i 's 2012 export share to destination j , measured before any country joined the system. The second term captures whether destination j belongs to cohort c and whether year t falls after that cohort joins the RTGS. Thus, $RTGS\ Exp_{i,t}$ varies across firms according to their pre-treatment destination mix and over time as additional countries join the system.

We estimate the following firm-level specification:

$$y_{i,t} = \exp \left\{ \beta_{RTGS} \cdot RTGS\ Exp_{i,t} + \beta_{Het} \cdot [RTGS\ Exp_{i,t} \times High\ EFD_{s,2012}] \right. \\ \left. + \alpha_i + \alpha_{s,t} + \sum_{\tau=2010}^{2019} \alpha_{TFP,\tau} \cdot [High\ TFP_{i,2012} \times \mathbb{I}\{t = \tau\}] + \beta_d \cdot Export\ Demand_{i,t} \right\} \cdot \epsilon_{i,t}. \quad (17)$$

The outcome $y_{i,t}$ is firm i 's total export volume in year t , aggregated across all destinations. The coefficient β_{RTGS} captures the average effect of RTGS exposure on total exports. The interaction coefficient β_{Het} tests the external-finance-dependence prediction. The variable $High\ EFD_{s,2012}$ equals one if firm i operates in a sector with above-median external financial dependence, measured as the ratio of trade credit payables to total sales in 2012. A positive β_{Het} indicates that firms in sectors with greater working-capital needs benefit more from payment integration.

The specification includes firm fixed effects, α_i , which absorb time-invariant firm characteristics, and sector-by-year fixed effects, $\alpha_{s,t}$, which absorb sector-specific shocks over time. We also allow high-productivity firms to follow flexible year-specific trends by interacting an indicator for above-median 2012 total factor productivity with year dummies.¹³ Finally, $Export\ Demand_{i,t}$ controls for time-varying demand conditions in each firm's destination markets. This variable

¹³We estimate firm TFP by assuming a Cobb–Douglas aggregate production function.

is constructed by weighting destination-specific demand shocks, described in Internet Appendix [IA.3](#), by the firm's 2012 export shares, analogously to the RTGS exposure measure in [Equation \(16\)](#).

[Table 8](#) presents the results. Column (1) shows a positive but statistically insignificant average effect of RTGS exposure on total exports when we do not account for firm heterogeneity. Column (2) reveals substantial heterogeneity by external financial dependence: the baseline effect for low-EFD firms is small and statistically insignificant, while firms in high-EFD sectors experience large and significant increases in total exports following payment-system integration. Columns (3)–(5) show that this pattern is not explained by other firm characteristics such as size, age, or total factor productivity. These results are consistent with the working-capital channel in the framework: faster, more reliable settlement reduces the financing burden of exporting, especially for firms whose production and sales are more exposed to external finance.

[INSERT [TABLE 8](#) ABOUT HERE]

[Table 9](#) provides complementary evidence using additional firm-level outcomes. Firms in sectors with high external financial dependence not only increase exports after RTGS exposure, but also experience increases in receivables, payables, and total sales. These outcomes are consistent with expanded trade activity and larger working-capital positions following payment integration. Through the lens of the framework, these outcomes are most informative in levels rather than as ratios: faster settlement can shorten the duration of outstanding claims even as trade scales up. The concentration of these effects among high-EFD firms further supports the interpretation that RTGS integration relaxes liquidity-related frictions.

[INSERT [TABLE 9](#) ABOUT HERE]

Taken together, these results indicate that payment integration has real effects by easing financial frictions. Faster and more reliable settlement narrows the gap between shipment and usable cash, allowing firms with greater working-capital needs to expand export activity. This evidence is closely related to [Barrot and Nanda \(2020\)](#), who show that faster payments alleviate financial

constraints for U.S. firms, and to the broader literature on financial frictions in international trade (Feenstra et al., 2014; Muûls, 2015; Niepmann and Schmidt-Eisenlohr, 2017b).

5.2.2 Bank Linkages and Exchange-Rate Risk

We next examine whether the baseline export response reflects two alternative or complementary channels: pre-existing bank linkages and exchange-rate risk. These tests help distinguish the payment-settlement channel from other mechanisms through which financial integration may affect trade.

First, the RTGS may have larger effects in destinations with weaker pre-existing banking links to South Africa. Recent evidence shows that access to well-connected banking networks affects the quality of cross-border payment routes: when direct connections are lost, banks rely on longer, more indirect, and more fragile payment chains (Matvos and Neiman, 2026). In our setting, a strong pre-existing presence of South African banks may similarly have provided shorter and more reliable payment routes before RTGS adoption, thereby reducing the marginal benefit of the new settlement platform. Conversely, where such banking links were weak, firms were more reliant on indirect settlement arrangements and therefore had more to gain from regional payment integration. The payment-settlement channel thus predicts larger RTGS effects in destinations with weaker pre-existing banking connectivity.

Second, RTGS integration could matter by reducing exchange-rate-risk exposure if faster settlement shortens the time during which firms are exposed to currency movements. Although this channel is plausible in principle, the SADC-RTGS settles rand-denominated transactions and primarily changes settlement speed, payment certainty, and interbank processing. The framework therefore predicts that the main effects should operate through transaction costs, settlement delays, and counterparty risk rather than through exchange-rate volatility.

In Table 10, we test these channels directly. We find that pre-existing banking integration, measured by South African banks' presence in destination countries, explains part of the baseline heterogeneity: effects are smaller in destinations with stronger banking links to

South Africa before RTGS adoption. This pattern is consistent with payment integration generating the largest gains where prior cross-border banking connectivity was more limited. By contrast, exchange-rate volatility has modest explanatory power. This evidence suggests that RTGS integration primarily operates by reducing payment-settlement frictions rather than by lowering exchange-rate-risk exposure.

[INSERT [TABLE 10](#) ABOUT HERE]

Overall, the mechanism tests point to a coherent interpretation. RTGS integration increases exports by reducing the settlement delay and interbank processing costs associated with cross-border settlement. These gains are strongest for firms whose working-capital needs make settlement delays more costly and in corridors where weak pre-existing banking links imply greater reliance on indirect payment routes ([Matvos and Neiman, 2026](#)). The evidence is less consistent with exchange-rate risk as the primary channel.

5.3 RTGS, Inter-bank Market and Correspondent Banking Amplification

The framework in [Section 3](#) implies that banks matter because they determine the settlement path available to exporting firms. Cross-border RTGS improves settlement reliability and speed for participating banks. Because adoption entails a fixed cost, our framework predicts that only banks with sufficiently large settlement volume in participating corridors join the system directly (Prediction 3). Adopter banks can then use the improved settlement technology not only for their own clients, but also to provide settlement services to non-adopting respondent banks through correspondent relationships. This section provides bank-level evidence on both margins. We first study which banks are more likely to adopt the system. We then test whether banks that are more likely to adopt expand interbank assets after the country in which they are located joins the RTGS. In the framework, such an expansion is consistent with adopter banks increasing correspondent settlement activity and transmitting the benefits of RTGS beyond direct participants.

We use information from all banking groups in the BankFocus database with at least one subsidiary in Africa. We first analyze which banks are more likely to join the payment system by 2019.¹⁴ Motivated by the adoption margin of Prediction 3, we expect banks with pre-existing cross-border group connections to RTGS countries to be more likely to adopt the system: these banks settle larger volumes in participating corridors, so the settlement-cost savings from direct adoption are more likely to justify its fixed cost. They are also better positioned to provide settlement services to affiliated or respondent banks. To test this prediction, we estimate the following model for banks located in participating countries:

$$RTGS\ Participant_{b,2019} = \delta_{exp} \cdot RTGS\ Exposure_b + \beta \cdot X_b + \gamma_r + \varepsilon_b. \quad (18)$$

Here b denotes the bank, and r is the country where the bank is located. $RTGS\ Exposure_b$ is a binary variable indicating whether the bank has another subsidiary or headquarters in a country that adopts the payment system. We include bank-level controls X_b , including whether the bank belongs to a multinational conglomerate, total assets, and the ratio of operating revenue to total assets. The coefficient of interest, δ_{exp} , captures whether banks with group-level exposure to RTGS-participating countries are more likely to adopt the system.

In the second part of the analysis, we exploit the fact that these exposed banks are more likely to adopt the payment system to examine whether adoption is associated with greater interbank activity. This test is motivated by Prediction 3 in [Section 3](#). If adopter banks provide RTGS settlement services to non-adopting respondent banks, then banks more likely to adopt should increase interbank assets after their country joins the system. We estimate the following stacked difference-in-differences specification:

$$y_{b,c,t} = \delta_{RTGS} \cdot \left[RTGS\ Exposure_{b,c} \times Post\ RTGS_{r,t} \right] + \gamma_{b,c} + \gamma_{h,c,t} + \gamma_{r,c,t} + \varepsilon_{b,c,t}. \quad (19)$$

¹⁴See [SADC Banking Association Newsletter](#), Issue 1, 2019.

Here b denotes the bank, r is the country where the bank is located, h is the banking conglomerate, and c is the treatment cohort. The variable $RTGS\ Exposure_{b,c}$ corresponds to the *HQ/Foreign Subsidiary RTGS Exposure* measure described in Table 6, indicating whether the bank's conglomerate has a presence in an RTGS-participating country. $Post\ RTGS_{r,t}$ is an indicator equal to one after country r , where bank b is located, joins the system. We include bank-cohort fixed effects ($\gamma_{b,c}$), bank-country-year-cohort fixed effects ($\gamma_{r,c,t}$), and conglomerate-year-cohort fixed effects ($\gamma_{h,c,t}$). This set of fixed effects absorbs time-invariant bank heterogeneity and time-varying unobserved heterogeneity at the country and conglomerate levels. The coefficient of interest, δ_{RTGS} , captures changes in interbank market activity for banks more likely to adopt the payment system relative to other banks in the same country that are less likely to adopt it.

In Table 6, we focus on the fourteen countries adopting the RTGS, yielding a sample of 119 banks. The table shows that the main determinant of system adoption is whether a bank's conglomerate has headquarters or another subsidiary in an RTGS-participating country. By contrast, belonging to a multinational conglomerate without a connection to another participating country does not significantly affect the probability of adoption. Bank size also predicts adoption, consistent with the adoption cutoff in the framework, but the economic magnitude of having a foreign subsidiary in a participating country is substantially larger, corresponding to an effect similar to a 10-standard-deviation increase in assets. These effects are not driven by unobservable country characteristics, such as differences in domestic financial development. Overall, the evidence indicates that pre-existing cross-border banking-group connections are central to explaining which banks adopt the settlement technology.

[INSERT TABLE 6 ABOUT HERE]

We next test whether banks more likely to adopt the RTGS expand interbank activity after implementation. We focus on interbank assets for two reasons. First, they are observed before and after implementation, allowing us to study changes around adoption. Second, they are closely linked to the correspondent-bank margin emphasized in the framework: when

non-participating banks rely on participant banks to settle RTGS transactions, these relationships should be reflected in greater interbank activity by participant banks. In [Table 7](#), we use the stacked difference-in-differences specification in [Equation \(19\)](#), focusing on heterogeneous effects for banks more likely to adopt the system. Banks less likely to participate do not experience changes in interbank market behavior. In contrast, banks more likely to participate experience a significant increase in interbank assets after the country in which they are located joins the system. These effects are not driven by conglomerate-level or country-level unobserved heterogeneity. The results are consistent with RTGS-participating banks expanding correspondent settlement activity for non-participating counterparts.

[INSERT [TABLE 7](#) ABOUT HERE]

[Figure 2](#) displays the dynamic effects of the cross-border payment system on banks more likely to participate in the RTGS. The figure provides reassuring evidence that the increase in interbank assets occurs after system adoption and is not driven by differential pre-existing trends in interbank market activity. Overall, the results in this section support the correspondent channel in the framework: banks with international group connections to participating countries are more likely to adopt the RTGS, and likely adopters expand interbank assets after implementation. This pattern suggests that improvements in settlement technology strengthen correspondent banking activity and help transmit the benefits of cross-border payment integration beyond direct adopters.¹⁵

[INSERT [FIGURE 2](#) ABOUT HERE]

6 Aggregate Evidence and Tariff Equivalents

Taken together, the firm-level and bank-level results paint a coherent picture: the RTGS system reduces payment settlement frictions, which in turn facilitates trade for financially constrained

¹⁵These results are robust to alternative transformations and specifications. Internet Appendix [Table IA.7](#) shows that our findings hold when using $\log(1 + y)$ transformations and PPML estimators, confirming that the observed effects are not sensitive to the specific functional form or estimation method.

firms. The bank-level evidence corroborates the firm-level findings by showing that the system operates through correspondent banking networks—banks with cross-border presence adopt the system and expand interbank activity, creating the infrastructure through which exporters benefit from faster payment settlement. We note, however, that faster payments may also affect trade through indirect channels—for instance, by reshaping financial networks or altering risk perceptions—that our bank-level evidence is consistent with but does not fully decompose. We now turn to aggregate evidence to assess the overall magnitude of these effects.

In addition to investigating firm-level consequences, we also assess the payment system's impacts on bilateral international trade flows using a standard gravity model. Gravity models are widely used in the international trade literature due to their strong theoretical foundations and empirical properties (Head and Mayer, 2014). To incorporate payment system participation into the standard gravity equation, we assume it affects international trade flows through bilateral trade costs. Therefore, we estimate:

$$X_{r,j,t} = \exp \left\{ \alpha + \beta_{RTGS} \cdot RTGS_{r,j,t} + \beta \cdot D_{r,j,t} + \gamma_{r,t} + \gamma_{j,t} + \varepsilon_{r,j,t} \right\} \quad (20)$$

$X_{r,j,t}$ represents the aggregated exports from country r to destination country j in year t . $RTGS_{r,j,t}$ is a binary variable that equals 1 when both countries r and j participate in the payment system in year t . We include exporter-by-year fixed effects and importer-by-year fixed effects. These are commonly used in gravity models to account for the “multilateral resistance” terms (Anderson and Van Wincoop, 2003). $D_{r,j,t}$ represents a set of bilateral control variables, including geographical distance and shared borders, commonly used in gravity estimations. We also include variables capturing dimensions of economic integration between countries, such as whether the countries have a free trade agreement or belong to the same customs union. We estimate.

We complement the findings of Equation (20) by estimating a similar specification when using alternative levels of aggregation of bilateral trade data. Specifically, we estimate:

$$X_{r,j,k,t} = \exp \left\{ \alpha + \beta_{RTGS} \cdot RTGS_{r,j,t} + \beta \cdot D_{r,j,t} + \gamma_{r,k,t} + \gamma_{j,k,t} + \varepsilon_{r,j,k,t} \right\} \quad (21)$$

$X_{r,j,k,t}$ represents the aggregated exports from country r to destination country j in a product or industry k , in year t . The variables $RTGS_{r,j,t}$ and $D_{r,j,t}$ are defined as in Equation (20). However, the more disaggregated data allow us to use more comprehensive fixed effects ($\gamma_{r,k,t}$ and $\gamma_{j,k,t}$), which further interact the country-by-time fixed effects with the product or industry k .

Table 11 presents the estimates of Equations (20) and (21) using the Poisson Pseudo-Maximum Likelihood estimator of Santos Silva and Tenreyro (2006). Column (1) uses the data at the *Origin Country* $\times$ *Destination Country* dimension. Column (2) uses the data at the *Origin Country* $\times$ *Destination Country* $\times$ *Product* dimension, with products classified according to the 2-digit Harmonized System (HS). Columns (3) and (4) use the data at the *Origin Country* $\times$ *Destination Country* $\times$ *Industry* dimension. Industries are defined according to the ISIC classification. To conduct the analysis at the industry level, we map the trade data from the 2-digit Harmonized System to the ISIC classification. To perform the cross-walk from products to industries, we use the concordance tables presented in Liao et al.'s (2020). Because some products can be matched to multiple industries, the results can be sensitive to the cross-walk process of mapping a product into multiple industries. To test the robustness of the results to the cross-walk, we conduct two alternative approaches. In column (3), we use the first industry matched to a product, and column (4) uses the last industry matched. Consistent with the firm-level results, Table 11 shows that the trade flows between participants of the payment system increase once countries join the system. Our results suggest an increase in such exports of 35% to 49% on average.

The country-level and firm-level estimates deliver remarkably similar results, with aggregate effects (35—49%) slightly exceeding our firm-level estimates (35—38.5%), revealing complementary insights from these two approaches. While our firm-level analysis provides tightly identified

causal effects by isolating intensive-margin adjustments of continuing exporters, it does not capture general equilibrium adjustments in response to payment system integrations. The larger aggregate estimates likely capture additional channels, including extensive margin adjustments through new firm entry and potential cross-firm spillovers.

To better illustrate the economic implications of these results, we can compute the *tariff equivalent* effect implied by these estimates, i.e., the equivalent *ad valorem* tariff rate that, if removed, would have had the same effect on trade as participation in the RTGS. Despite not including tariffs in our specification, we can compute the *tariff equivalent* effect of the RTGS by relying on (i) the structural gravity model properties and (ii) trade elasticity estimations from the literature (Yotov et al., 2016).¹⁶ In the last three rows of Table 11, we compute the RTGS tariff-equivalent effects using different estimates of the trade elasticity from the literature. Using the aggregate elasticity estimate from Caliendo and Parro (2015), we find that participating in the RTGS has an equivalent effect on trade as eliminating a tariff rate of 6.8%–9.2%. Alternatively, using the trade elasticity estimate from Simonovska and Waugh (2014), the tariff-equivalent effect is 7.8%–10.5%, while when using the elasticity estimate from Baier and Bergstrand (2007), we obtain rates between 6.2% and 8.3%. To put these magnitudes in context, the simple average MFN applied tariff rate across SADC member states is approximately 10%, ranging from under 1% in Mauritius to nearly 19% in Zimbabwe, while the Southern African Customs Union average stands at 8.5%.¹⁷ These tariff equivalents thus imply that payment system integration generates trade gains on the order of eliminating a substantial share of existing tariffs in the region. These magnitudes are also comparable to previous studies that estimate the impact of time delays in export procedures. For example, Hummels and Schaur (2013) shows that each day in transit is equivalent to an *ad valorem* tariff of 0.6–2.1%, implying that reducing payment settlement time from 7 days to real-time could generate tariff-equivalent effects of 4–14%—close to our calculated tariff equivalent of 6.2–10.5 percentage points.

¹⁶The tariff equivalent effect can be calculated by $\left[e^{\hat{\beta}_{RTGS}/(\hat{\sigma})} - 1 \right] \times 100$, where $\hat{\sigma}$ is the trade elasticity from the literature.

¹⁷Source: WTO Tariff and Trade Data (ttd.wto.org), various years (2020–2025); WTO, *Trade Policy Review: Southern African Customs Union*, WT/TPR/S/447, 2023.

[INSERT [TABLE 11](#) ABOUT HERE]

In sum, the estimates presented in [Table 11](#) confirm the substantial effect of participating in the RTGS on trade flows. The estimates from the gravity equations are similar but slightly larger than the coefficients obtained from the firm-level analysis in [Table 3](#). To further gauge the dynamic aggregate effects of the RTGS, we also run an aggregate event-study specification. [Figure 3](#) shows that exports from participating countries followed a similar trend in the years preceding the implementation, but they began to diverge after the countries joined the system. Remarkably, we find the dynamic effects to be very similar to those at the firm level in [Figure 1](#). These results support our empirical strategy and suggest that the effects persist over time.

Overall, the aggregate analysis validates and extends our firm-level findings, demonstrating that payment system integration generates substantial trade gains comparable to a 6.2–10.5% tariff reduction. The convergence between our firm-level and country-level estimates provides robust evidence that cross-border payment integration generates significant impacts on international trade. This consistency across methodological approaches underscores that modernizing and integrating payment systems across countries can deliver trade benefits on par with major trade liberalization agreements, offering policymakers an alternative lever for promoting international trade.

[INSERT [FIGURE 3](#) ABOUT HERE]

7 Empirical Validity Tests

In this section, we present a series of results that test the validity of our identification strategy by demonstrating sharp dynamic effects, consistency of the underlying mechanism, and quasi-random treatment assignment.

The following tests support our identification strategy by exploiting the timing and nature of the RTGS implementation, examining heterogeneous responses consistent with our proposed mechanisms, and assessing the quasi-experimental nature of treatment timing.

7.1 High-Frequency Analysis: Sharpness of the Effects

In this section, we employ a high-frequency empirical strategy to assess the immediate effects of the RTGS implementation on firms' exports to participating countries. By focusing on narrow time windows around the system implementation date, this approach enables us to detect sudden changes in export behavior that are closely tied to the RTGS going live, thereby strengthening our identification. In particular, the sharpness and timing of the export response provide compelling evidence that our results are driven by the payment system's implementation itself, rather than by confounding factors or broader, slower-moving processes such as regional trade integration or macroeconomic trends, which would be expected to generate more gradual and diffuse effects.

In [Table IA.1](#), we exploit a temporal regression discontinuity design (RD) centered on the implementation date of the RTGS system in each trade partner country. This design uses time as the forcing variable and adopts a within-firm perspective, comparing export outcomes in a short window before and after the policy change (e.g., [Davis, 2008](#)). By holding firm characteristics constant over this narrow time horizon, we reduce the influence of time-varying confounders and isolate the causal impact of introducing the RTGS. The RD approach exploits the quasi-experimental nature of the implementation timing and the high-frequency data structure to deliver a clean identification of the treatment effect.

[INSERT [TABLE IA.1](#) ABOUT HERE]

Using optimal bandwidths, we find that the introduction of the RTGS led to a 21.8% statistically significant increase in export volumes and a 15.4% rise in the number of export transactions in the months immediately following implementation. These discontinuous jumps in the outcome variables around the threshold reinforce the view that the effects we document are directly attributable to the RTGS adoption, underscoring the causal interpretation of our results.

7.2 Trade Margin Differentiation: Intensive vs. Extensive Margins

Our main findings demonstrate a sharp increase in intensive trade margins—export volume and transaction counts—shortly after RTGS adoption. In contrast, we observe no statistically or economically significant change in extensive trade margins over the same short-run window. This pattern is presented in [Table IA.2](#), which reports results for three outcomes: the number of export destinations (Columns 1–2), the number of destination-by-product combinations at the HS 4-digit level (Columns 3–4), and destination-by-product combinations at the HS 6-digit level (Columns 5–6). Across all specifications—and even when interacting RTGS exposure with high external financial dependence—the estimated coefficients remain small in magnitude and statistically insignificant.

The absence of extensive margin effects provides important insights into the mechanisms through which payment system integration operates. Our null results for the number of export destinations and product-market combinations suggest that the RTGS system does not meaningfully reduce the fixed costs or informational barriers that impede firms from entering new markets, at least not within the time frame of our analysis. This pattern aligns with empirical evidence from [Paravisini, Rappoport, Schnabl, and Wolfenzon \(2015\)](#), who documents that credit shocks have strong and immediate effects on export volumes, but no impact on firms’ entry into new markets or products.

[INSERT [TABLE IA.2](#) ABOUT HERE]

Taken together, our findings indicate that faster and more reliable cross-border payments primarily alleviate variable trade costs related to working capital and payment frictions, thereby facilitating more frequent and higher-volume transactions. However, they may not be sufficient to overcome the structural barriers that prevent firms from expanding into new destinations or products. This pattern reinforces our interpretation that the causal effect of the RTGS implementation drives the observed short-run trade response. If our results were driven by confounding factors

such as broader regional economic integration, improved diplomatic relations, or general business environment reforms, we would expect simultaneous changes in both intensive and extensive margins. The fact that only intensive margins respond while extensive margins remain unchanged suggests that our estimates capture the specific impact of cross-border payment system integration rather than spurious correlations with other unobserved trends affecting trade more broadly.

7.3 Predicting the Timing of RTGS Adoption

In Internet Appendix [Table IA.3](#) and [Table IA.4](#), we examine whether observable pair- and country-specific characteristics predict the timing of RTGS adoption. We regress indicator variables for treatment in 2013 and 2014 on standardized measures—including lagged bilateral trade, geographic distance, time-zone differences, GDP per capita, ease-of-doing-business metrics—with destination fixed effects. The comparison group in each specification consists of pairs treated in later years.

The results indicate that these pre-determined characteristics do not systematically predict treatment timing. The lone exception is a marginally significant coefficient on origin trade flows in 2014; however, this isolated finding does not suggest a broader pattern. When we focus specifically on South African trade partners ([Table IA.4](#)), we find similar results: no characteristic significantly predicts timing, and the overall explanatory power (R^2) remains low.

These findings contribute to internal validity by alleviating concerns about endogenous timing. If treatment timing were driven by observable factors, it could introduce confounding. However, the lack of predictive power supports the identification assumption that timing is “as good as random” conditional on destination fixed effects. Empirical best practices for causal inference in staggered treatment settings underscore the importance of such verification steps. Within difference-in-differences and event-study frameworks, establishing that pre-determined covariates are not correlated with treatment timing strengthens the validity of the analysis and supports the assumption of parallel trends across groups.

In sum, the null results from our timing regressions indicate that neither trade intensity nor geographic or institutional characteristics appear to drive RTGS entry timing. This finding strengthens our causal claims: the observed changes in trade volumes and transactions are unlikely to reflect selection into treatment based on pre-existing conditions.

8 Conclusion

Exploiting confidential administrative data on South African exporters, we show that cross-border payment-system integration has large real effects on international trade. Firms increase export volumes by about 35% and transaction frequency by about 45% after destination countries join the RTGS system. At the aggregate level, the increase in trade between participating countries is comparable to a 6–8 percentage-point tariff reduction. These gains reflect trade creation rather than trade diversion: firms exposed to RTGS destinations do not reduce exports to non-participating markets.

Our results identify payment settlement as a distinct financial friction in international trade. Cross-border transactions depend not only on credit availability, but also on the infrastructure through which payments are routed, settled, and made available as usable liquidity. Consistent with this mechanism, the export response is concentrated in destinations with pre-existing domestic RTGS infrastructure, where regional integration can generate end-to-end speed gains. The effects are also stronger among firms in sectors with high external financial dependence, for which delays and uncertainty in payment settlement impose greater working-capital costs. These firms also experience increases in total sales, receivables, and payables, consistent with higher trade activity and working-capital positions after payment integration.

Payment integration also affects financial intermediation itself. Banks with greater ex ante propensity to adopt the RTGS expand interbank assets following system activation, consistent with increased correspondent settlement activity. In our framework, banks with access to the regional settlement platform can provide settlement services to respondent banks that do not

participate directly. Payment integration can therefore deepen interbank markets and transmit improvements in settlement technology beyond direct participants. This financial-development effect complements the firm-level evidence and shows that payment infrastructure shapes both real activity and the organization of cross-border banking.

These mechanisms have broader relevance beyond the SADC setting. Recent evidence shows that geopolitical shocks and financial sanctions can disrupt banks' access to international payment networks and force transactions onto longer, more indirect, and more fragile payment chains ([Matvos and Neiman, 2026](#)). Our results highlight the other side of this relationship: expanding regional settlement capacity can reduce dependence on indirect payment routes, strengthen regional interbank links, and support trade. Regional payment integration may therefore provide an important source of resilience for emerging markets when access to global payment networks becomes more uncertain.

More importantly, our findings show that financial integration is not only about capital flows, credit supply, or foreign-bank presence. The infrastructure through which banks route and settle cross-border payments is itself a crucial dimension of financial integration. Investments in regional payment systems can reduce settlement frictions, relax working-capital constraints, deepen interbank markets, and facilitate international trade. As geopolitical risk places greater pressure on the architecture of global payments, understanding the real and financial effects of such infrastructure becomes increasingly important.

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Figures

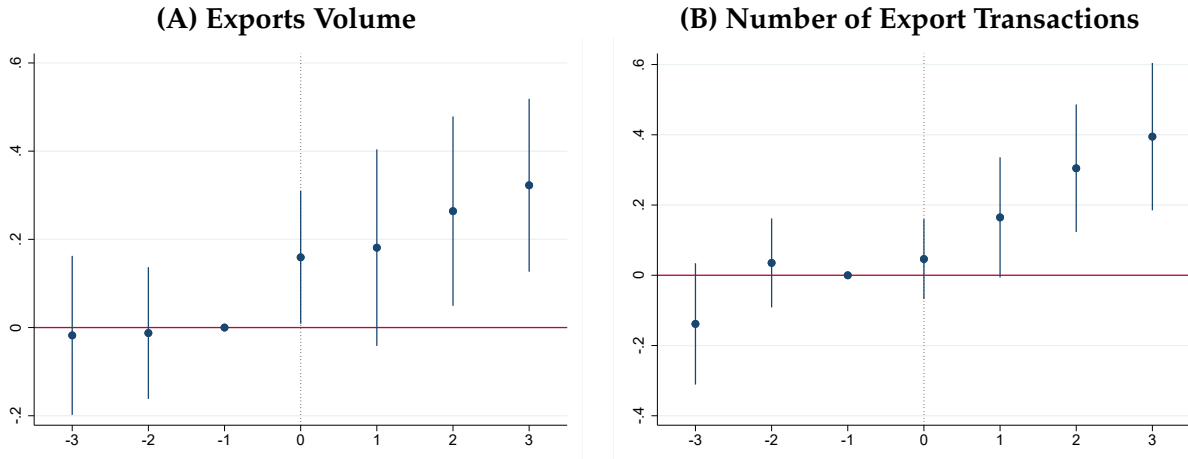


Figure 1. Firm-Level Exports around the Payment System Integration. This figure shows the dynamic effects of the implementation of the Real-Time Gross Settlement (RTGS) system on firm-destination export volumes (Panel A) and the number of transactions (Panel B). The event-study coefficients are estimated relative to the year before the RTGS implementation ($t - 1$). Reported 95% confidence intervals are based on standard errors clustered at the *Firm* $\times$ *Destination Country* $\times$ *Cohort* and the *Firm* $\times$ *Date* $\times$ *Cohort* levels. All specifications include *Firm* $\times$ *Destination Country* $\times$ *Cohort* and *Year* $\times$ *Continent* $\times$ *Cohort* fixed effects, as well as time-varying trade partner export demand. Estimation uses the Poisson Pseudo-Maximum Likelihood (PPML) method of Santos Silva and Tenreiro (2006).

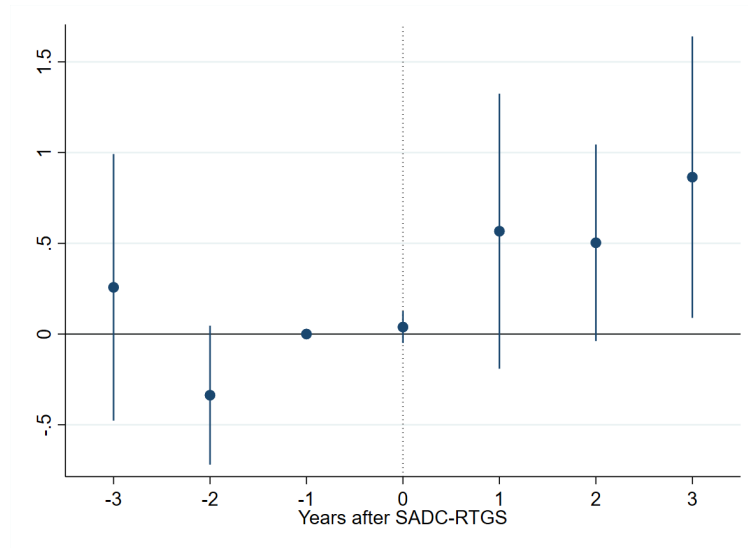


Figure 2. The Response of Interbank Assets to Payment System Integration. This figure shows the evolution of interbank assets following the implementation of the Real-Time Gross Settlement (RTGS) system. Estimates are relative to 2012, one year before the initial implementation in July 2013. Reported 95% confidence intervals are based on standard errors clustered at the bank-cohort level. The figure displays the dynamic specification of Column 3 from [Table 7](#). The dependent variables use inverse hyperbolic sine transformation. The specification includes *Origin Bank* $\times$ *Cohort*, *Country* $\times$ *Cohort* $\times$ *Year*, and *Conglomerate* $\times$ *Cohort* $\times$ *Year* fixed effects.

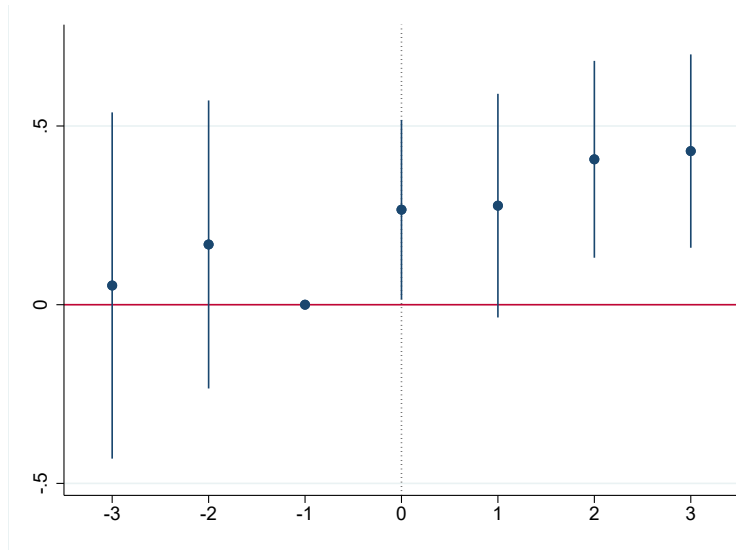


Figure 3. Country-Level Bilateral Trade Flows and Cross-Border Payment System Integration. This figure shows the dynamic effects of the Real-Time Gross Settlement (RTGS) system participation on bilateral trade flows between member countries. Estimates are relative to one year before the initial implementation in July 2013. Reported 95% confidence intervals are based on standard errors clustered at the *Origin Country* $\times$ *Destination Country* level. All specifications include *Origin Country* $\times$ *Cohort* and *Time* $\times$ *Cohort* fixed effects, as well as pair-specific controls (Southern African Customs Union, Southern African Development Community Free Trade Area, Free Trade Agreement, Customs Unions, geographical contiguity indicators, and log(distance)). Estimation uses the Poisson Pseudo-Maximum Likelihood (PPML) method of Santos Silva and Teneyro (2006).

Tables

Table 1. Real-Time Gross Settlement System: Implementation Timeline. This table presents the chronological development of the Southern African Development Community's (SADC) Real-Time Gross Settlement (RTGS) system from 2010 to 2016. It shows the key milestones in the approval process and the dates when different member countries joined the system. The implementation followed a phased approach, beginning with countries in the Common Monetary Area (CMA) in July 2013 and expanding to include all fourteen participating countries by November 2016.

Date	Event
October 2010	SADC Committee of Central Bank Governors (CCBG) approves undertaking the proof of concept in the CMA region.
April 2013	CCBG approves the legal agreements. CMA Governors sign the agreements.
July 2013	SADC-RTGS goes live in Lesotho, Namibia, South Africa and Swaziland.
April 2014	SADC-RTGS goes live in Malawi, Tanzania, and Zimbabwe.
September 2014	SADC-RTGS goes live in Mauritius and Zambia.
July 2016	SADC-RTGS goes live in Seychelles.
October 2016	SADC-RTGS goes live in Angola, Botswana, and Mozambique.
November 2016	SADC-RTGS goes live in The Democratic Republic of Congo.

Table 2. Summary Statistics: Firm-Level Matched Sample. This table reports summary statistics for firms in our matched sample, constructed using Coarsened Exact Matching on firm sector, size, and total factor productivity (TFP). The treated group consists of firms exporting to countries that joined the payment system, while the control group includes comparable firms exporting only to non-participating countries. All variables are measured at baseline (i.e., $t - 1$), one year before the Southern African Development Community's Real-Time Gross Settlement (RTGS) inclusion. Monetary values are in millions of South African rand (ZAR), deflated to 2012 prices using the Consumer Price Index (CPI). The p -values in column 4 are adjusted for multiple-hypothesis testing following [Clarke, Romano, and Wolf \(2020\)](#) with 5,000 repetitions.

	All Firms (1)	Treated Firms (2)	Control Firms (3)	Mean Difference Test p -value (4)
Loans	0.130	0.128	0.135	0.610
Receivables	1.060	0.547	1.936	0.610
Payables	0.598	0.639	0.528	0.341
Capital Stock	0.579	0.486	0.736	0.348
Value Added	1.123	1.174	1.035	0.376
Exports to Sales Ratio	0.040	0.037	0.046	0.230
Exports	0.248	0.193	0.341	0.312
Observations	52,694	33,199	19,495	

Table 3. Baseline Results: Firm-Destination Exports and Cross-Border Payment Integration. This table examines the effects of the Real-Time Gross Settlement (RTGS) system on firm-destination export outcomes. Columns 1-2 present results for export volume, while columns 3-4 show effects on the number of export transactions. All regressions use a matched sample constructed via Coarsened Exact Matching on firm sector and sales. Standard errors are clustered at the *Firm* $\times$ *Destination Country* $\times$ *Cohort* and *Firm* $\times$ *Year* $\times$ *Cohort* levels. Estimation uses the Poisson Pseudo-Maximum Likelihood (PPML) method of Santos Silva and Tenreyro (2006). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Exports Volume		Exports Transactions	
	(1)	(2)	(3)	(4)
Post-RTGS	0.326*** (0.082)	0.300*** (0.079)	0.385*** (0.078)	0.374*** (0.076)
Export Demand	0.643*** (0.123)	0.637*** (0.123)	0.493*** (0.110)	0.483*** (0.105)
Firm $\times$ Cohort FE	✓		✓	
Destination Country $\times$ Cohort FE	✓		✓	
Firm $\times$ Destination Country $\times$ Cohort FE		✓		✓
Continent $\times$ Cohort $\times$ Year FE	✓	✓	✓	✓
Observations	1,507,450	1,507,389	1,507,552	1,507,491
R^2	0.643	0.873	0.742	0.876

Table 4. Spillover Analysis: Effects on Non-RTGS Destinations. This table examines whether the Real-Time Gross Settlement (RTGS) system implementation affects exports to non-participating countries. The variable *RTGS Firm Spillover* captures potential effects on exports to non-RTGS countries by firms that also export to RTGS-participating countries. Columns (1) and (2) present results for export volume, while columns (3) and (4) show effects on the number of export transactions. Standard errors are clustered at the *Firm × Destination Country × Cohort* and *Firm × Year × Cohort* levels. Estimation uses Santos Silva and Tenreiro's (2006) Poisson Pseudo-Maximum Likelihood (PPML) method. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Exports Volume		Exports Transactions	
	(1)	(2)	(3)	(4)
Post-RTGS	0.392*** (0.090)	0.386*** (0.095)	0.357*** (0.088)	0.384*** (0.087)
RTGS Firm Spillover	0.094* (0.052)	0.122* (0.071)	-0.039 (0.062)	0.013 (0.061)
Export Demand	0.646*** (0.123)	0.644*** (0.122)	0.493*** (0.110)	0.483*** (0.105)
Firm × Cohort FE	✓		✓	
Destination Country × Cohort FE	✓		✓	
Firm × Destination Country × Cohort FE		✓		✓
Continent × Cohort × Year FE	✓	✓	✓	✓
Observations	1,507,450	1,507,389	1,507,552	1,507,491
R^2	0.643	0.873	0.742	0.876

Table 5. Firm-Destination Exports and RTGS Integration: The Role of Fast Payments. This table examines how domestic Real-Time Gross Settlement (RTGS) adoption interacts with cross-border RTGS participation to affect exports. The variable *Post-RTGS* indicates when a destination country joins the RTGS system, while *Domestic RTGS* is a binary variable indicating whether the destination country had implemented its own domestic real-time gross settlement system. *RTGS Firm Spillover* captures potential effects on exports to non-RTGS countries by firms that also export to RTGS-participating countries. *Export Demand* controls for destination-country specific demand conditions. Standard errors are clustered at the *Firm* $\times$ *Destination Country* $\times$ *Cohort* and *Firm* $\times$ *Year* $\times$ *Cohort* level. Estimation uses the Poisson Pseudo-Maximum Likelihood (PPML) method of Santos Silva and Tenreiro (2006). Information on domestic RTGS adoption dates comes from D'Andrea and Limodio (2024). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Exports Volume		Exports Transactions	
	(1)	(2)	(3)	(4)
Post-RTGS	-0.083 (0.176)	0.008 (0.178)	-0.376** (0.186)	-0.232 (0.171)
$\times$ Domestic RTGS	0.481** (0.193)	0.383** (0.190)	0.746*** (0.220)	0.627*** (0.209)
RTGS Firm Spillover	0.094* (0.052)	0.122* (0.071)	-0.039 (0.062)	0.013 (0.061)
Export Demand	0.657*** (0.125)	0.653*** (0.124)	0.538*** (0.117)	0.521*** (0.113)
Firm $\times$ Cohort FE	✓		✓	
Destination Country $\times$ Cohort FE	✓		✓	
Firm $\times$ Destination Country $\times$ Cohort FE		✓		✓
Continent $\times$ Cohort $\times$ Year FE	✓	✓	✓	✓
Observations	1,507,389	1,507,389	1,507,491	1,507,491
R^2	0.643	0.873	0.742	0.876

Table 6. RTGS Bank Adoption. Standard errors are clustered at the *Bank* and *Country* levels. A multinational conglomerate is a dummy variable that indicates if the conglomerate has at least one subsidiary in a country other than the headquarter (HQ) country. HQ/Foreign Subsidiary RTGS Exposure is a dummy variable indicating that the conglomerate has at least one subsidiary or the HQ in a country that adopts RTGS at time t . Measures of total assets and operating revenues to asset ratio are standardized. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	SADC-RTGS Participant in 2019			
	(1)	(2)	(3)	(4)
HQ/Foreign Subsidiary RTGS Exposure	0.361** (0.143)	0.361** (0.157)	0.382** (0.154)	0.458*** (0.144)
Multinational Conglomerate	-0.131 (0.148)	0.196 (0.128)		
Total Assets				0.041** (0.019)
$\frac{\text{Operating Revenues}}{\text{Total Assets}}$				0.020 (0.020)
Country FE		✓		
Country $\times$ Multinational Cong. FE			✓	✓
Observations	119	119	118	102
R ²	0.111	0.401	0.445	0.519

Table 7. RTGS Bank Adoption and Interbank Markets. The dependent variables are the inverse hyperbolic sine transformation of the original variables. Standard errors are clustered at the *Bank* $\times$ *Cohort* level. A multinational *Conglomerate* is defined as a binary variable that indicates if the conglomerate has at least one subsidiary in a country other than the HQ country. HQ/Foreign Subsidiary RTGS Exposure is a dummy variable indicating that the conglomerate has at least one subsidiary or the HQ in a country that adopted the RTGS at time t . Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Interbank Assets T. Assets			Interbank Liabilities T. Assets		
	(1)	(2)	(3)	(4)	(5)	(6)
Post-RTGS at Country	-0.010 (0.042)			-0.063* (0.037)		
Post-RTGS at Country $\times$ HQ/Foreign Subsidiary RTGS Exposure	0.206*** (0.079)	0.297** (0.137)	0.503** (0.248)	0.028 (0.068)	0.005 (0.094)	0.039 (0.160)
Bank $\times$ Cohort FE	✓	✓	✓	✓	✓	✓
HQ Country $\times$ Cohort $\times$ Year FE	✓	✓		✓	✓	
Country $\times$ Cohort $\times$ Year FE		✓	✓		✓	✓
Conglomerate $\times$ Cohort $\times$ Year FE			✓			✓
Observations	13,046	12,374	11,174	12,114	11,432	10,306
R ²	0.481	0.544	0.606	0.474	0.541	0.595

Table 8. Firms' Total Exports and RTGS Integration: Heterogeneous Effects by External Financial Dependence. This table analyzes how a firm's exposure to the RTGS system affects its total exports, with a focus on sectoral financial constraints. The variable *RTGS Exp* is the exposure variable defined in Equation (16), measuring the percentage of a firm's 2012 exports going to countries that participated in the RTGS system in year t . *High EFD* represents sectors with above-median external financial dependence, measured as the ratio of payables (trade credit) to total sales in 2012. The interactions with firm characteristics examine heterogeneous effects: *Big* represents firms with above-mean sales in 2012, *Older* represents firms with above-median age in 2012, *High TFP* indicates firms with above-median total factor productivity in 2012, while *Primary* and *Services* are sector classification dummies. Standard errors are clustered at the $Firm \times Sector \times Year$ level. Estimation uses the Poisson Pseudo-Maximum Likelihood (PPML) method of Santos Silva and Tenreiro (2006). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Total Exports				
	(1)	(2)	(3)	(4)	(5)
RTGS Exposure	0.045 (0.076)	-0.170* (0.104)	-0.197* (0.114)	-0.163 (0.113)	-0.183* (0.104)
× High EFD		0.449*** (0.140)	0.413*** (0.144)	0.403*** (0.142)	0.445*** (0.144)
× Big			-0.003 (0.077)		
× Older				0.083 (0.076)	
× High TFP					0.272* (0.148)
Export Demand	0.073 (0.116)	0.041 (0.116)	0.041 (0.116)	0.046 (0.115)	0.041 (0.116)
Firm FE	✓	✓	✓	✓	✓
TFP × Year FE	✓	✓	✓	✓	✓
Sector × Year FE	✓	✓	✓	✓	✓
Observations	40,748	40,748	40,748	40,748	40,748
R^2	0.926	0.926	0.924	0.924	0.924

Table 9. Firms' Additional Outcomes and RTGS Integration Outcome variables include: Total Sales (gross sales revenue), Value Added (gross sales minus cost of sales), Receivables (net trade and other receivables after provisions), Payables (trade and other payables), and Labor Costs (wages including medical benefits, pensions, and direct remuneration). The variable *RTGS Exp* is the exposure variable defined in Equation (16), measuring the percentage of a firm's 2012 exports going to countries that participated in the RTGS system in year t . *High EFD* represents sectors with above-median external financial dependence, measured as the ratio of payables (trade credit) to total sales. Standard errors are clustered at the Firm $\times$ Destination Country $\times$ Cohort and Firm $\times$ Year $\times$ Cohort level. Regressions use the matched sample from the Coarsened Exact Matching procedure using the firm sector and firm sales. Our specification uses Santos Silva and Tenreiro's (2006) Poisson Pseudo-Maximum Likelihood Estimator (PPML). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Total Sales		Value Added		Receivables		Payables		Labor Costs	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
RTGS Exposure	-0.025 (0.041)	-0.107 (0.068)	0.022 (0.029)	0.002 (0.034)	-0.017 (0.043)	-0.132** (0.059)	0.071* (0.043)	-0.033 (0.043)	-0.000 (0.026)	0.002 (0.029)
RTGS Exposure $\times$ High EFD		0.168** (0.078)		0.038 (0.054)		0.214*** (0.078)		0.186** (0.074)		-0.005 (0.051)
Export Demand	-0.000 (0.054)	-0.002 (0.053)	0.107* (0.063)	0.106* (0.063)	0.046 (0.076)	0.046 (0.075)	0.083 (0.079)	0.081 (0.078)	0.027 (0.040)	0.027 (0.040)
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
TFP $\times$ Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	40,684	40,684	40,293	40,293	39,209	39,209	40,023	40,023	39,618	39,618
R^2	0.950	0.950	0.934	0.934	0.888	0.888	0.918	0.918	0.926	0.926

Table 10. Firm-Destination Exports and RTGS Integration: Mechanisms. Standard errors clustered at the $Firm \times Destination$ Country $\times Cohort$ and $Firm \times Year \times Cohort$ level. Regressions use the matched sample from the Coarsened Exact Matching procedure using the firm sector and firm sales. Our specification uses Santos Silva and Tenreiro's (2006) Poisson Pseudo-Maximum Likelihood Estimator (PPML). Information from the date of adoption of the Domestic RTGS comes from D'Andrea and Limodio (2024). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Exports Volume			Exports Transactions		
	(1)	(2)	(3)	(4)	(5)	(6)
Post-RTGS	0.021 (0.178)	-0.005 (0.179)	0.020 (0.178)	-0.256 (0.174)	-0.220 (0.174)	-0.241 (0.175)
× Domestic RTGS	0.419** (0.196)	0.388** (0.191)	0.425** (0.198)	0.693*** (0.232)	0.619*** (0.212)	0.699*** (0.233)
× Banking Integration	-0.395* (0.219)		-0.407* (0.223)	-0.397** (0.170)		-0.409** (0.171)
× Exc. Rates Vol.		-0.001 (0.005)	-0.003 (0.005)		-0.008 (0.006)	-0.010* (0.006)
RTGS Firm Spillover	0.118 (0.078)	0.122* (0.071)	0.119 (0.079)	-0.033 (0.071)	0.005 (0.060)	-0.038 (0.071)
× Banking Integration	0.033 (0.184)		0.031 (0.184)	0.296* (0.157)		0.303* (0.158)
× Exc. Rates Vol.		-0.000 (0.001)	-0.000 (0.001)		0.001 (0.001)	0.001 (0.001)
Export Demand	0.676*** (0.129)	0.662*** (0.129)	0.681*** (0.130)	0.599*** (0.145)	0.507*** (0.119)	0.602*** (0.146)
Firm × Destination Country × Cohort FE	✓	✓	✓	✓	✓	✓
Continent × Cohort × Year FE	✓	✓	✓	✓	✓	✓
Observations	1,457,158	1,457,158	1,457,158	1,457,260	1,457,260	1,457,260
R ²	0.873	0.873	0.873	0.877	0.877	0.877

Table 11. Country-Level Evidence: Gravity Estimations. Standard errors in parentheses are clustered at the level of the respective fixed effects. Our specification uses Santos Silva and Tenreyro's (2006) Poisson Pseudo-Maximum Likelihood (PPML) Estimator. Control variables include the geographical distance between countries, contiguity, whether countries have a free trade agreement, and whether they belong to the SADC customs union or other customs unions. Column (1) uses the data at the *Origin Country* $\times$ *Destination Country* dimension, column (2) uses the data at the *Origin Country* $\times$ *Destination Country* $\times$ *Product* dimension, with products classified according the 2-digit Harmonized System (HS). Columns (3) and (4) use the data at the *Origin Country* $\times$ *Destination Country* $\times$ *Industry* dimension. Industries are defined according to the ISIC classification. To conduct the analysis at the industry level, we perform a mapping of the trade data from the 2-digit Harmonized System to ISIC classification. Columns (3) and (4) represent the alternative approaches for mapping of products into industries. The last three rows compute the tariff equivalent effect of the SADC-RTGS using different trade elasticity estimates from the literature. The tariff equivalent effects are calculated using $\left[e^{\hat{\beta}_{RTGS} / (\hat{\sigma})} - 1 \right] \times 100$. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Bilateral Trade Flows			
	Country Level	Country-Product Level	Country-Industry Level	
	(1)	(2)	(3)	(4)
<i>RTGS</i> $\times$ <i>Post</i>	0.399** (0.163)	0.384*** (0.128)	0.331*** (0.126)	0.299** (0.119)
Control Variables	✓	✓	✓	✓
Origin Country $\times$ Year FE	✓			
Destination Country $\times$ Year FE	✓			
Origin Country $\times$ Product $\times$ Year FE		✓		
Destination Country $\times$ Product $\times$ Year FE		✓		
Origin Country $\times$ Industry $\times$ Year FE			✓	✓
Destination Country $\times$ Industry $\times$ Year FE			✓	✓
Observations	355,320	29,396,253	7,916,271	7,697,008
Post RTGS Tariff Equivalent				
$\hat{\sigma} = 4$ (Simonovska and Waugh, 2014)	10.5%	10.1%	8.6%	7.8%
$\hat{\sigma} = 4.55$ (Caliendo and Parro, 2015)	9.2%	8.8%	7.5%	6.8%
$\hat{\sigma} = 5$ (Baier and Bergstrand, 2007)	8.3%	8.0%	6.8%	6.2%

Internet Appendix

“Unleashing International Trade through Financial Integration: Evidence from a Cross-Border Payment System”

NOT FOR PUBLICATION

IA.1 Conceptual Framework Details

This appendix provides the derivations and proofs for Section 3. The framework has three components. First, exporters face a payment-settlement friction when selling abroad. We take the exporter-bank relationship as predetermined at the time of RTGS adoption, so exporters do not costlessly choose the lowest-cost settlement provider in each corridor. Second, cross-border RTGS affects this friction through settlement delays and counterparty risk, while bank-corridor settlement costs capture persistent heterogeneity in banks' ability to process payments in a given corridor; these bank-corridor costs are invariant to RTGS access. Third, when a country joins the RTGS, each bank weighs the fixed cost of direct adoption against settling through an adopter bank at a per-transaction fee. Banks with sufficiently large RTGS-relevant settlement volume adopt; the remaining banks route their cross-border payments through adopter banks, which expands correspondent settlement activity and, through cost pass-through, transmits the friction reduction to the exporters they serve.

IA.1.1 Main Assumptions and Notation

Throughout the appendix, we maintain the following assumptions: (i) the demand elasticity satisfies $\sigma > 1$; (ii) sectoral external financial dependence is denoted by $d_s \in (0,1)$; (iii) in each corridor j , RTGS access reduces the settlement-delay component if and only if the destination has domestic RTGS infrastructure, so the speed gain $\lambda_j \equiv T_j(0, D_j) - T_j(1, D_j)$ satisfies $\lambda_j > 0$ if $D_j = 1$ and $\lambda_j = 0$ if $D_j = 0$, and RTGS access weakly reduces counterparty risk, so the risk reduction $\eta_j \equiv \delta_j(0) - \delta_j(1)$ satisfies $\eta_j \geq 0$; (iv) RTGS access is binary, $z \in \{0,1\}$, and comparative statics in z are therefore stated as discrete differences rather than derivatives; (v) every component of the payment-settlement friction is nonnegative — $T_j(z, D_j) \geq 0$ and $\delta_j(z) \geq 0$ for all (z, D_j) , bank-corridor costs $c \geq 0$, and correspondent fees $\phi \geq 0$ — so the friction satisfies $\omega \geq 0$ under every settlement route considered and $1 + d_s \omega > 0$ always holds; (vi) the fixed cost of RTGS adoption satisfies $F > 0$.

IA.1.2 Exporter Problem

Consider firm i in sector s exporting to destination j . Let $b(i)$ denote firm i 's pre-existing bank at the time of RTGS adoption. We take the firm-bank relationship as predetermined, capturing the stickiness of banking relationships and the bundled nature of cross-border banking services, and consistent with evidence that bank-specific conditions transmit to the exports of the firms matched to those banks (Amiti and Weinstein, 2011; Paravisini et al., 2015). Exporters typically obtain payment execution, cash management, foreign-exchange services, compliance screening, documentation, and credit from an existing banking relationship rather than continuously re-optimizing over all possible banks for each destination. The relevant payment-settlement friction for firm i exporting to destination j is therefore the friction associated with its pre-existing bank, $\omega_{b(i)j}$.¹⁸

Demand is CES with elasticity $\sigma > 1$:

$$q_{ijs} = A_j p_{ijs}^{-\sigma}. \quad (22)$$

A fraction $d_s \in (0,1)$ of variable export costs is exposed to settlement frictions, capturing the sector's external financial dependence. The exporter's effective marginal cost through its pre-existing bank is

$$C_{ijs}^e = C_{ijs}(1 + d_s \omega_{b(i)j}). \quad (23)$$

The exporter solves

$$\max_{p_{ijs}} \left\{ p_{ijs} q_{ijs} - C_{ijs}(1 + d_s \omega_{b(i)j}) q_{ijs} - f_{js} \right\}, \quad \text{s.t.} \quad q_{ijs} = A_j p_{ijs}^{-\sigma}. \quad (24)$$

¹⁸Equivalently, the framework can be interpreted as one in which firms choose among banks subject to relationship-specific switching costs. Firm i selects the bank that minimizes $\omega_{bj} + \kappa_{ib}$, where κ_{ib} captures the cost of establishing or moving a banking relationship. When these switching costs are sufficiently large, the relevant settlement friction is determined by the firm's pre-existing bank, $b(i)$. This formulation delivers the same comparative statics while avoiding the implication that firms costlessly select the lowest-cost settlement provider in every corridor.

The first-order condition yields the standard CES markup:

$$p_{ijs}^* = \frac{\sigma}{\sigma - 1} C_{ijs} (1 + d_s \omega_{b(i)j}). \quad (25)$$

Substituting into the revenue expression gives

$$X_{ijs} = A_j \left[\frac{\sigma - 1}{\sigma} \frac{1}{C_{ijs} (1 + d_s \omega_{b(i)j})} \right]^{\sigma - 1}. \quad (26)$$

The payment-settlement friction $\omega_{b(i)j}$ is a continuous variable, so exports may be differentiated with respect to it. Taking logs and differentiating:

$$\frac{\partial \log X_{ijs}}{\partial \omega_{b(i)j}} = - \frac{(\sigma - 1) d_s}{1 + d_s \omega_{b(i)j}} < 0, \quad (27)$$

where the strict inequality uses $\sigma > 1$, $d_s > 0$, and $1 + d_s \omega_{b(i)j} > 0$ from the maintained assumptions. Exports are strictly decreasing in the payment-settlement friction.

IA.1.3 The Payment-Settlement Friction and RTGS Access

The payment-settlement friction faced by firm i in destination j decomposes as

$$\omega_{b(i)j}(z) = T_j(z, D_j) + \delta_j(z) + c_{b(i)j}. \quad (28)$$

Here, $T_j(z, D_j)$ captures settlement delays, $\delta_j(z)$ captures counterparty risk, and $c_{b(i)j}$ captures persistent bank-corridor settlement costs. The indicator $z \in \{0, 1\}$ denotes whether firm i 's pre-existing bank has access to RTGS settlement in corridor j , and $D_j \in \{0, 1\}$ indicates whether destination j has domestic RTGS infrastructure. As in Section 3, we impose no functional forms on T_j and δ_j ; the maintained assumptions restrict only the discrete differences

$$\lambda_j \equiv T_j(0, D_j) - T_j(1, D_j), \quad \eta_j \equiv \delta_j(0) - \delta_j(1), \quad (29)$$

with $\lambda_j > 0$ if $D_j = 1$, $\lambda_j = 0$ if $D_j = 0$, and $\eta_j \geq 0$. End-to-end speed gains require both RTGS access by the bank ($z = 1$) and domestic RTGS infrastructure in the destination ($D_j = 1$), while the counterparty-risk reduction requires only RTGS access.

Bank-corridor costs $c_{b(i)j}$ capture heterogeneity in banks' ability to process cross-border transactions in a given corridor, including operational capacity, correspondent networks, compliance systems, treasury operations, and corridor-specific experience. These costs are a characteristic of the bank-corridor pair and are invariant to RTGS access: RTGS adoption changes the settlement technology available to the firm's pre-existing bank, holding fixed both the firm-bank match and the bank-corridor cost component $c_{b(i)j}$. Because firm-bank relationships are sticky, this heterogeneity generates variation in the payment-settlement friction faced by otherwise similar exporters.

Since z is binary, the effect of RTGS access on the friction is a discrete difference. Substituting (29) into (28) gives

$$\Delta_z \omega_{b(i)j} \equiv \omega_{b(i)j}(1) - \omega_{b(i)j}(0) = -(\lambda_j + \eta_j) \equiv -\Gamma_j \leq 0, \quad (30)$$

with strict inequality if and only if at least one channel is active in corridor j , that is, if and only if $D_j = 1$ (so that the speed channel operates, given $\lambda_j > 0$ in that case) or $\eta_j > 0$ (so that the counterparty-risk channel operates). Combining (30) with the export equation (26), and writing $\omega_{b(i)j}^0 \equiv \omega_{b(i)j}(0)$ for the baseline (no-access) friction, the export response to RTGS access is the discrete difference

$$\Delta_z \log X_{ijs} \equiv \log X_{ijs}(\omega_{b(i)j}(1)) - \log X_{ijs}(\omega_{b(i)j}(0)) = (\sigma - 1) \log \left[\frac{1 + d_s \omega_{b(i)j}^0}{1 + d_s (\omega_{b(i)j}^0 - \Gamma_j)} \right] \geq 0, \quad (31)$$

with strict inequality if and only if $\Gamma_j > 0$. Expression (31) makes explicit that the export response depends on the baseline friction $\omega_{b(i)j}^0$ at which the discrete change is evaluated; the cross-corridor comparisons below therefore hold the baseline friction fixed explicitly.

IA.1.4 Proof of Prediction 1

The prediction compares the export response to RTGS access across destinations with and without domestic RTGS infrastructure. Because the response (31) depends on the baseline friction, we state the *ceteris paribus* condition explicitly.

Assumption 1 (*Ceteris Paribus* Comparison Across Destinations). The comparison is between two destinations j and j' that are identical in all respects relevant for the export response — in particular, they share the same baseline payment-settlement friction, $\omega_{b(i)j}^0 = \omega_{b(i)j'}^0 \equiv \omega^0$, and the same counterparty-risk reduction, $\eta_j = \eta_{j'} \equiv \eta$ — and differ only in domestic RTGS infrastructure: $D_j = 1$ and $D_{j'} = 0$.

For the destination with domestic RTGS infrastructure ($D_j = 1$), both the end-to-end speed channel and the counterparty-risk channel operate, so, writing $\lambda \equiv \lambda_j > 0$ by the maintained assumptions, $\Gamma_j = \lambda + \eta > 0$, and

$$\Delta_z \log X_{ijs} \Big|_{D_j=1} = (\sigma - 1) \log \left[\frac{1 + d_s \omega^0}{1 + d_s (\omega^0 - \lambda - \eta)} \right] > 0. \quad (32)$$

For the destination without domestic RTGS infrastructure ($D_{j'} = 0$), the domestic payment leg remains a bottleneck for end-to-end settlement speed and $\lambda_{j'} = 0$ by the maintained assumptions, so only the counterparty-risk channel operates, $\Gamma_{j'} = \eta$, and

$$\Delta_z \log X_{ij's} \Big|_{D_{j'}=0} = (\sigma - 1) \log \left[\frac{1 + d_s \omega^0}{1 + d_s (\omega^0 - \eta)} \right] \geq 0, \quad (33)$$

with strict inequality if and only if $\eta > 0$. Taking the difference of (32) and (33), the terms in $\log(1 + d_s \omega^0)$ cancel under Assumption 1, leaving

$$\Delta_z \log X_{ijs} \Big|_{D_j=1} - \Delta_z \log X_{ij's} \Big|_{D_{j'}=0} = (\sigma - 1) \log \left[\frac{1 + d_s (\omega^0 - \eta)}{1 + d_s (\omega^0 - \lambda - \eta)} \right] > 0, \quad (34)$$

where the strict inequality follows because $\lambda > 0$ implies $\omega^0 - \lambda - \eta < \omega^0 - \eta$, both denominators are strictly positive by the maintained assumptions, and log is strictly increasing. Thus, RTGS access weakly increases exports, strictly so when the destination has domestic RTGS infrastructure, and the effect is strictly larger when domestic RTGS infrastructure enables end-to-end speed gains. $\square$

IA.1.5 Proof of Prediction 2

Fix a corridor j with baseline friction $\omega_{b(i)j}^0$ and let $\Gamma_j = \lambda_j + \eta_j \geq 0$ as in (30). Define the export response as a function of external financial dependence,

$$g(d_s) \equiv \Delta_z \log X_{ijs} = (\sigma - 1) \left[\log(1 + d_s \omega_{b(i)j}^0) - \log(1 + d_s(\omega_{b(i)j}^0 - \Gamma_j)) \right]. \quad (35)$$

Since d_s is continuous, differentiate with respect to d_s :

$$g'(d_s) = (\sigma - 1) \left[\frac{\omega_{b(i)j}^0}{1 + d_s \omega_{b(i)j}^0} - \frac{\omega_{b(i)j}^0 - \Gamma_j}{1 + d_s(\omega_{b(i)j}^0 - \Gamma_j)} \right]. \quad (36)$$

The function $h(\omega) \equiv \omega / (1 + d_s \omega)$ satisfies

$$h'(\omega) = \frac{1}{(1 + d_s \omega)^2} > 0, \quad (37)$$

so h is strictly increasing on the relevant domain. Because $\omega_{b(i)j}^0 > \omega_{b(i)j}^0 - \Gamma_j$ whenever $\Gamma_j > 0$, it follows that $g'(d_s) > 0$ whenever $\Gamma_j > 0$, and $g'(d_s) = 0$ when $\Gamma_j = 0$. RTGS effects are therefore stronger for firms with higher external financial dependence whenever RTGS reduces at least one component of the payment-settlement friction in corridor j . $\square$

IA.1.6 Proof of Prediction 3

IA.1.6.1 Environment

When a country joins the RTGS, each bank b operating there chooses one of three settlement arrangements for its RTGS-relevant settlement volume $V_b > 0$, defined as the volume of cross-border payments the bank settles in corridors covered by the system: adopt the system directly (A), settle through an adopter bank as a correspondent (C), or retain the non-RTGS status quo (N). To ease notation, we suppress corridor heterogeneity and write $\Gamma > 0$ for the per-transaction friction reduction delivered by RTGS rails; all statements below hold corridor-by-corridor with Γ_j in place of Γ in every corridor with $\Gamma_j > 0$. In corridors with $\Gamma_j = 0$, RTGS rails deliver no friction reduction, no fee satisfies Assumption 2, and banks retain the status quo for that corridor's volume; such corridors contribute nothing to the adoption decision and are excluded from the RTGS-relevant settlement volume V_b .

Direct adoption requires a fixed cost $F > 0$ per period, capturing participation and connectivity requirements, compliance systems, and the treasury operations needed to maintain pre-funded balances; it does not scale with settlement volume. The correspondent route carries a per-transaction fee ϕ paid to the adopter bank. Normalizing the status quo payoff to zero, bank b 's settlement-cost savings under each arrangement are

$$\Pi_b(A) = \Gamma V_b - F, \quad \Pi_b(C) = (\Gamma - \phi) V_b, \quad \Pi_b(N) = 0. \quad (38)$$

Assumption 2 (Fee Bounds). The correspondent fee satisfies $0 < \phi < \Gamma$.

The bounds are participation constraints. A fee below the friction reduction Γ is required for any bank to use the correspondent route. Any strictly positive fee covers the adopter's marginal cost of processing additional payments on rails it already operates, which we normalize to zero, so adopters supply correspondent services at any $\phi > 0$. Where the fee falls within $(0, \Gamma)$ is determined by competition among adopter banks; the results below hold for every fee in this interval.

Assumption 3 (Cost Pass-Through to Exporters). Banks price settlement services to their exporter clients at marginal cost, so the payment-settlement friction faced by an exporter equals the friction of the settlement arrangement chosen by its bank.

Assumption 3 makes the transmission mechanism explicit: absent this assumption, a bank with market power over its exporter clients could absorb part of the friction reduction in its own margins.

IA.1.6.2 Adoption Cutoff

Lemma 1 (Adoption Cutoff). *Under Assumption 2, bank b adopts if and only if*

$$V_b \geq V^* \equiv \frac{F}{\phi}, \quad (39)$$

banks with $V_b < V^$ choose the correspondent route, and no bank retains the status quo.*

Proof. By Assumption 2 and $V_b > 0$, $\Pi_b(C) = (\Gamma - \phi)V_b > 0 = \Pi_b(N)$, so the correspondent route strictly dominates the status quo for every bank. Comparing adoption with the correspondent route,

$$\Pi_b(A) \geq \Pi_b(C) \iff \Gamma V_b - F \geq (\Gamma - \phi)V_b \iff \phi V_b \geq F \iff V_b \geq \frac{F}{\phi}, \quad (40)$$

□

Remark 1 (Fee Income). Adopter banks also earn fee income ϕ per transaction on the correspondent volume they attract, which raises the value of adoption for banks expecting correspondent business and weakly lowers their effective cutoff. Incorporating this term changes no qualitative statement below, so we omit it.

IA.1.6.3 Interbank Settlement Expansion

Proposition 1 (Interbank Settlement Expansion). *Suppose the distribution of RTGS-relevant settlement volume straddles the cutoff: some banks have $V_b \geq V^*$ and some have $V_b < V^*$. Then after RTGS*

activation, adopter banks settle, in addition to their own volume, the correspondent volume

$$\sum_{b: V_b < V^*} V_b > 0 \quad (41)$$

on behalf of non-adopting banks. Interbank (correspondent) settlement activity therefore expands relative to the pre-RTGS baseline, in which each bank settled its own volume through the non-RTGS route.

Proof. Immediate from Lemma 1: every bank below the cutoff strictly prefers the correspondent route to both alternatives and routes its RTGS-relevant volume through an adopter bank. Before activation, no such correspondent volume existed on RTGS rails. $\square$

IA.1.6.4 Transmission to Exporters at Non-Adopting Banks

Corollary 1 (Transmission Beyond Direct Adopters). *Under Assumptions 2 and 3, an exporter served by a non-adopting bank faces friction $\omega^0 - (\Gamma - \phi) < \omega^0$, where ω^0 is the friction under the pre-RTGS status quo, and by (27) its exports strictly rise.*

Proof. By Lemma 1, a non-adopting bank chooses the correspondent route, which lowers its per-transaction settlement cost by $\Gamma - \phi > 0$ relative to the status quo. By Assumption 3, its exporter clients face the correspondingly lower friction, and by (27) exports are strictly decreasing in the friction. $\square$

Remark 2 (Magnitude of the Transmission). The friction reduction transmitted to exporters at non-adopting banks equals $\Gamma - \phi \in (0, \Gamma)$ and is an equilibrium object. Competition among adopter banks for correspondent business pushes ϕ toward marginal cost, enlarging the transmission, while a correspondent with market power prices close to Γ , shrinking it. Note from Lemma 1 that a lower fee also raises the cutoff $V^* = F/\phi$, so stronger competition simultaneously enlarges the transmission and shifts more banks into the correspondent route. The sign of the transmission is pinned down by Corollary 1, while its magnitude is an empirical question.

IA.1.6.5 Endogenous Settlement Volume

The baseline analysis treats bank b 's RTGS-relevant settlement volume V_b as invariant to its choice of settlement arrangement. However, by equation (27), a lower payment-settlement friction raises the exports of the bank's clients, and the friction reduction is strictly larger under direct adoption (Γ) than under the correspondent route ($\Gamma - \phi$). The volume a bank settles should therefore itself respond to the arrangement it chooses, and be larger under direct adoption. This subsection relaxes the invariance assumption and shows that all results carry over.

Let $V_b > 0$ now denote bank b 's settlement volume under the correspondent (C) route, and suppose direct adoption (A) scales this volume by a factor γ :

$$V_b(A) = \gamma V_b, \quad V_b(C) = V_b, \quad \gamma \geq 1, \quad (42)$$

where $\gamma \geq 1$ captures, in reduced form, the settlement business induced by the larger friction reduction on RTGS rails — the export growth of the bank's own clients implied by (27). Correspondent volume attracted by direct access is compensated at the fee ϕ rather than the cost saving Γ and is therefore not part of γ ; as in Remark 1, incorporating that fee income would only reinforce adoption incentives. Banks anticipate the volume response when choosing their arrangement, so the payoffs in (38) become

$$\Pi_b(A) = \gamma \Gamma V_b - F, \quad \Pi_b(C) = (\Gamma - \phi) V_b, \quad \Pi_b(N) = 0. \quad (43)$$

The baseline model is nested at $\gamma = 1$.

Proposition 2 (Adoption Cutoff with Endogenous Volume). *Under Assumption 2 and (42), bank b adopts if and only if*

$$V_b \geq V_\gamma^* \equiv \frac{F}{\phi + (\gamma - 1)\Gamma}, \quad (44)$$

banks with $V_b < V_\gamma^*$ choose the correspondent route, and no bank retains the status quo. The cutoff satisfies $V_\gamma^* \leq V^* = F/\phi$, with equality if and only if $\gamma = 1$, and is strictly decreasing in γ .

Proof. As in Lemma 1, $\Pi_b(C) = (\Gamma - \phi)V_b > 0 = \Pi_b(N)$ by Assumption 2 and $V_b > 0$, so the correspondent route strictly dominates the status quo for every bank. Comparing adoption with the correspondent route,

$$\Pi_b(A) \geq \Pi_b(C) \iff \gamma\Gamma V_b - F \geq (\Gamma - \phi)V_b \iff [\phi + (\gamma - 1)\Gamma]V_b \geq F \iff V_b \geq \frac{F}{\phi + (\gamma - 1)\Gamma}, \quad (45)$$

The denominator satisfies $\phi + (\gamma - 1)\Gamma \geq \phi > 0$ since $\gamma \geq 1$ and $\Gamma > 0$. The cutoff comparison and monotonicity follow directly: $\phi + (\gamma - 1)\Gamma \geq \phi$ with equality if and only if $\gamma = 1$, and $\partial V_\gamma^* / \partial \gamma = -F\Gamma / [\phi + (\gamma - 1)\Gamma]^2 < 0$. $\square$

Adoption remains a cutoff rule in V_b , so every result in the baseline analysis carries over with V_γ^* in place of V^* : selection into adoption by settlement volume is unchanged, Proposition 1 holds provided the volume distribution straddles V_γ^* , and Corollary 1, which does not depend on volume, is unaffected. The only effect of endogenizing volume is to lower the adoption cutoff: because the friction reduction is larger on RTGS rails, banks internalize the induced expansion of their settlement business, and direct adoption becomes profitable for banks that would not adopt at fixed volume. Endogenous volume therefore amplifies adoption and, if anything, strengthens the selection mechanism behind Prediction 3.

IA.1.6.6 Empirical Implication

Prediction 3 implies two bank-level patterns. First, selection into adoption: banks with larger RTGS-relevant settlement volume — in practice, banks with headquarters or foreign subsidiaries in participating countries — are more likely to adopt the system. Second, interbank expansion: adopter banks settle on behalf of non-adopting banks and should therefore expand correspondent settlement activity after the country in which they operate joins the system. In the empirical

analysis, we proxy correspondent settlement activity using interbank assets. The bank-level prediction is that banks more likely to adopt the RTGS experience a relative increase in interbank assets after the country in which they operate joins the system. Evidence of such an increase is consistent with correspondent banking propagating the benefits of settlement integration beyond direct participants and with payment integration deepening interbank markets.

IA.2 Additional Results

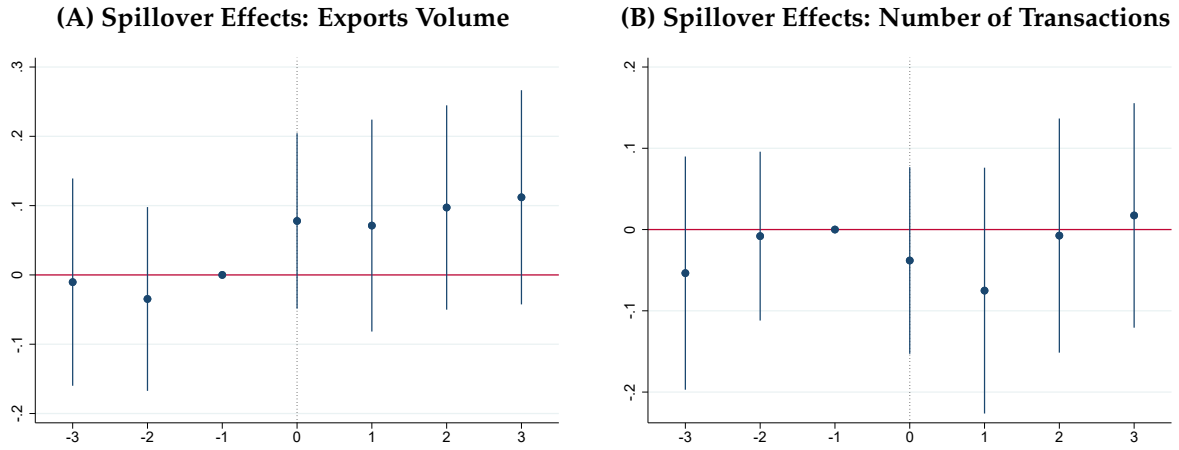


Figure IA.1. Firm-Destination Exports and RTGS Integration. Reported 90% confidence intervals are based on standard errors clustered at the $Firm \times Destination \times Cohort$ and $Firm \times Year \times Cohort$ level. All specifications include $Firm \times Destination \times Cohort$ and $Year \times Continent \times Cohort$ fixed effects as well as time-varying trade partner export demand. Our specification uses Santos Silva and Tenreiro's (2006) Poisson Pseudo-Maximum Likelihood (PPML) Estimator.

Table IA.1. High-Frequency Identification using Monthly Firm-Destination Exports: Regression Discontinuity Design. Standard errors clustered at the $Firm \times Destination\ Country \times Cohort$ level. Our specification uses Santos Silva and Tenreiro's (2006) Poisson Pseudo-Maximum Likelihood (PPML) Estimator. The local linear regression includes a dummy for post-RTGS implementation, time since the cutoff date, squared value of time since the cutoff date, and the interactions between these two variables and the Post-RTGS implementation dummy. Each pair of columns estimates separate regressions for the different windows around the implementation date represented in months, as indicated in their title. The controls include demand export and $Firm \times Destination\ Country$ fixed effects. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

	Bandwidth (-2,2)		Bandwidth (-3,3)		Bandwidth (-4,4)		Bandwidth (-5,5)	
					Opt. Band. Exp. Vol.		Opt. Band. Exp. Trans.	
	Exp. Volume	Exp. Trans.	Exp. Volume	Exp. Trans.	Exp. Volume	Exp. Trans.	Exp. Volume	Exp. Trans.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post-RTGS	0.114* (0.066)	0.173*** (0.041)	0.127 (0.144)	0.279*** (0.100)	0.197** (0.094)	0.258*** (0.041)	0.115* (0.066)	0.143*** (0.030)
Firm $\times$ Destination FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	92,267	92,267	143,030	143,030	192,187	192,187	246,929	246,929
R ²	0.842	0.847	0.825	0.841	0.821	0.838	0.811	0.837

Table IA.2. Effects of the RTGS Integration on Firms' Export Destinations and Products. Standard errors clustered at the *Firm* $\times$ *Sector* $\times$ *Year* level. *RTGS Exp* is the exposure variable defined in Equation (16). Regressions use the matched sample of the Coarsened Exact Matching procedure using the firm sector and firm sales. Our specification uses Santos Silva and Tenreiro's (2006) Poisson Pseudo-Maximum Likelihood (PPML) Estimator. High EFD represents sectors with above the median payables (trade credit) to total sales. Columns 1 and 2 show the estimates using the number of destination countries as the dependent variable. Columns 3 and 4 use the destination-number of products at the 6-digit (HS6) level of the Harmonized System classification, while Columns 5 and 6 use the destination-number of the more aggregated 4-digit (HS4) level products as the dependent variable. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Number of Destinations		Destination 4-digit HS Products		Destination 6-digit HS Products	
	(1)	(2)	(3)	(4)	(5)	(6)
RTGS Exp	-0.015 (0.019)	-0.014 (0.020)	0.024 (0.029)	0.060 (0.037)	0.032 (0.031)	0.070* (0.040)
$\times$ High EFD		-0.002 (0.036)		-0.077 (0.051)		-0.082 (0.056)
Export Demand	-0.006 (0.034)	-0.006 (0.034)	0.093 (0.059)	0.095 (0.059)	0.121* (0.067)	0.123* (0.067)
Firm FE	✓	✓	✓	✓	✓	✓
TFP $\times$ Year FE	✓	✓	✓	✓	✓	✓
Sector $\times$ Year FE	✓	✓	✓	✓	✓	✓
Observations	40,895	40,895	40,866	40,866	40,866	40,866
R^2	0.308	0.308	0.730	0.730	0.780	0.780

Table IA.3. Predictors of RTGS Treatment Timing. Robust standard errors are shown in parentheses. All predictors are the standardized original variable at the entry year. The control group for each regression is comprised of trade pairs treated afterward. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

	SADC-RTGS in 2013	SADC-RTGS in 2014
	(1)	(2)
Pair-Specific Characteristics		
Previous Year Bilateral Trade flows (1,000 Current USD)	-0.903 (0.918)	3.364 (2.230)
Time Difference (hours)	0.048 (0.168)	0.144 (0.368)
Distance (1,000 km)	-0.264 (0.211)	0.312 (0.338)
Origin-Specific Characteristics		
Origin Time to start a business (days)	0.016 (0.028)	0.054 (0.047)
Origin GDP, PPP (1,000 Current USD)	0.328 (0.416)	0.744 (0.461)
Origin Trade flows (1,000 Current USD)	1.372 (1.243)	-4.251* (2.560)
Destination FE	✓	✓
Observations	133	121
R ²	0.309	0.519

Table IA.4. Predictors of RTGS timing: South African Trade Partners. Robust standard errors in parentheses. Predictors are the standardized original variable at the entry year. The control group for each regression is comprised of trade pairs treated afterward. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

	SADC-RTGS in 2013	SADC-RTGS in 2014
	(1)	(2)
One year lag bilateral trade	-0.415 (0.962)	-2.225 (1.466)
Time Difference (hours)	0.959 (1.310)	-0.188 (3.166)
Distance (1,000 km)	-1.665 (1.394)	-1.662 (3.599)
Destination's time required to start a business (days)	0.109 (0.187)	-0.057 (0.305)
Observations	13	10
R ²	0.281	0.289

Table IA.5. Firm-Destination Exports and RTGS Integration: Non-Matched Sample. Standard errors clustered at the *Firm* $\times$ *Destination Country* $\times$ *Cohort* and *Firm* $\times$ *Year* $\times$ *Cohort* level. Our specification uses Santos Silva and Tenreiro's (2006) Poisson Pseudo-Maximum Likelihood (PPML) Estimator. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Exports Volume		Exports Transactions	
	(1)	(2)	(3)	(4)
Post-RTGS	0.316*** (0.078)	0.332*** (0.080)	0.379*** (0.077)	0.354*** (0.081)
Export Demand	0.606*** (0.125)	0.600*** (0.125)	0.453*** (0.105)	0.433*** (0.113)
Firm $\times$ Cohort FE	✓		✓	
Destination Country $\times$ Cohort FE	✓		✓	
Firm $\times$ Destination Country $\times$ Cohort FE		✓		✓
Cohort $\times$ Year FE	✓	✓	✓	✓
Observations	1,562,916	1,562,978	1,563,020	1,563,082
R^2	0.883	0.672	0.875	0.743

Table IA.6. Firm-Destination Exports and RTGS Integration: Only African Trade Partners as Controls. Standard errors clustered at the *Firm* $\times$ *Destination* $\times$ *Cohort* and *Firm* $\times$ *Year* $\times$ *Cohort* level. Regressions use the matched sample of the Coarsened Exact Matching procedure using the firm sector and firm sales. Our specification uses Santos Silva and Tenreiro's (2006) Poisson Pseudo-Maximum Likelihood (PPML) estimator. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Exports Volume		Exports Transactions	
	(1)	(2)	(3)	(4)
Post-RTGS	0.223*** (0.079)	0.220*** (0.080)	0.384*** (0.077)	0.381*** (0.078)
Export Demand	0.176* (0.103)	0.178* (0.104)	0.469*** (0.122)	0.477*** (0.118)
Firm $\times$ Cohort FE	✓		✓	
Destination Country $\times$ Cohort FE	✓		✓	
Firm $\times$ Destination Country $\times$ Cohort FE		✓		✓
Cohort $\times$ Year FE	✓	✓	✓	✓
Observations	707,374	707,318	707,409	707,353
R^2	0.663	0.813	0.812	0.890

Table IA.7. Robustness: RTGS Bank Adoption and Interbank Markets. This table shows that our main findings from Table 7 are robust to alternative model specifications and estimators. Columns 1–3 report OLS estimates using the logarithm of interbank assets over total assets (plus one) as the dependent variable. Columns 4–6 use the Poisson Pseudo-Maximum Likelihood (PPML) estimator (Santos Silva and Tenreyro, 2006), with the interbank assets over total assets as the dependent variable. Standard errors are clustered at the bank-cohort level. A multinational *Conglomerate* is a dummy variable that indicates if the conglomerate has at least one subsidiary in a country other than the HQ country. HQ/Foreign Subsidiary RTGS Exposure is a dummy variable indicating that the conglomerate has at least one subsidiary or is headquartered in a country that adopts RTGS at time t . Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	$\log(1 + \frac{\text{Interbank Assets}}{T. \text{ Assets}})$			$\frac{\text{Interbank Assets}}{T. \text{ Assets}}$		
	(1)	(2)	(3)	(4)	(5)	(6)
Post-RTGS at Country	-0.009 (0.035)			-0.290 (0.260)		
Post-RTGS at Country × HQ/Foreign Subsidiary RTGS Exposure	0.161** (0.063)	0.241** (0.112)	0.411** (0.208)	1.085*** (0.396)	1.655** (0.712)	1.431 (1.217)
Bank × Cohort FE	✓	✓	✓	✓	✓	✓
HQ Country × Year × Cohort FE	✓	✓		✓	✓	
Country × Year × Cohort FE		✓	✓		✓	✓
Conglomerate × Year × Cohort FE			✓			✓
PPML Specification				✓	✓	✓
Observations	13,046	12,374	11,174	12,959	12,287	11,119

IA.3 Export Demand

Despite the empirical design and the inclusion of the control variables described in [Section 5](#), there remains concern that our estimates may be contaminated by additional country-specific shocks relevant to both our dependent variables and our measure of exposure to the RTGS system. One potential issue would be trade-partner-specific demand shocks that are also correlated with the country's participation in the SADC–RTGS.

We address this potential confounding factor and estimate a destination-specific component that captures year-by-year aggregated shocks to the country's demand for global exports. The idea is that this measure can capture the overall changes in the country's demand for global exports, excluding trade with South Africa. Following a strategy similar to [Costa, Garred, and Pessoa \(2016\)](#), we estimate an auxiliary regression using aggregated bilateral trade data for 2010–2019 (as discussed in [Section 4](#)) and excluding South Africa. Specifically, the estimated model is:

$$X_{i,j,t} = \alpha + \sum_{i,t} \delta_{i,t} \cdot \left[\mathbb{1}\{\text{Importer} = i\} \times \mathbb{1}\{\text{Year} = t\} \right] + \gamma_{i,j} + \epsilon_{i,j,t},$$

where $X_{i,j,t}$ is the volume of trade between importer i , and exporter j in year t . $\gamma_{i,j}$ represents a set of country-pair dummy variables that capture the time-invariant pair-specific determinants of bilateral trade, such as distance, common language, and contiguity. Our coefficient of interest is $\delta_{i,t}$, obtained from the interaction between an importer-specific dummy and a year-specific dummy variable. This means that the importer-year-specific coefficients represent the average deviations in bilateral trade of that country's imports in any given year, excluding trade with South Africa. We refer to the estimates $\hat{\delta}_{i,t}$ as the export-demand component and include them as control variables in our specifications to account for destination-specific demand shocks.